

A Random Walk through Probabilistic Graph Theory

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A thesis submitted to the Department of Mathematics
in partial fulfillment of the requirements for the
undergraduate honors program

Stanford University

June 2024

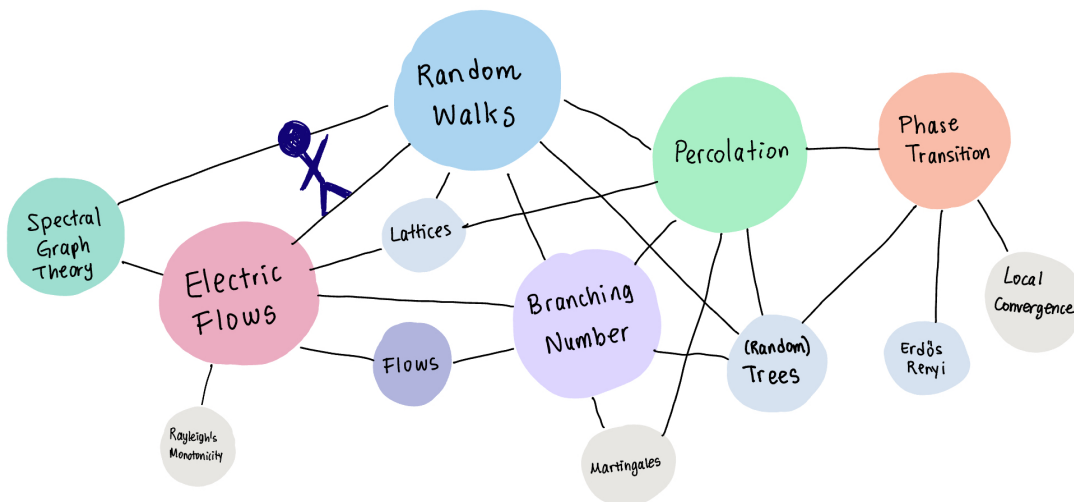
1. ACKNOWLEDGMENTS

This thesis would not have been possible without the support and guidance of my advisor, Prof. Tselil Schramm. I thank her for her patience in answering all of my questions and giving feedback on my writing, for bringing up interesting examples and connections to other areas, for providing the necessary intuition and resources as I studied the materials presented here, and for making time for me whenever I needed it beyond the scope of this thesis. I have learned a lot from her. I also would like to thank Prof. Amin Saberi and Yeganeh Alimohammadi for introducing me to the local convergence of graphs and small worlds, formulating exciting research questions, and supporting and mentoring me in my research journey. Lastly, I want to thank my family for always being there for me.

2. INTRODUCTION

We start by introducing the marvelous connection between electric networks and random walks on graphs, which allows us to view graphs as electric circuits and voltages as probabilities that a particle performing a random walk will reach the source node before the sink node(s). These concepts will eventually help us to classify random walks on infinite lattices and deterministic as well as random trees (!) as recurrent or transient walks, dealing with whether a person performing a random walk will return to its starting position almost surely. On the way, we will draw a lot of graphs, make a bunch of connections to other areas of mathematics, and introduce many different tools such as the theory of local convergence or the branching number, which is the reciprocal of the critical percolation threshold $p_c(T)$ of a tree T . This threshold p_c corresponds to the supremum of probabilities p such that if one thins edges of a rooted graph with probability $1 - p$, the resulting connected component that includes the root will have infinite size. We conclude with different random graph models, such as small-world networks¹ and Erdős-Rényi, that locally behave like random trees, which is why their threshold function for the emergence of the giant components is connected to the critical percolation threshold of the trees they resemble locally.

¹We give a novel proof for local convergence of small-world networks.



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3. ELECTRIC NETWORKS AND RANDOM WALKS

We will denote a *graph* by $G = (V(G), E(G))$ where $V := V(G)$ is the vertex or node set, and $E := E(G)$ is the edge set. G can be finite or infinite!

For this thesis, we will assume that G is connected and locally finite.² We will assign each undirected edge (x, y) a *resistance* R_{xy} , or equivalently, a *conductance* $C_{xy} = \frac{1}{R_{xy}}$. Next, we define a Markov chain with transition matrix P given by

$$P_{xy} = \frac{C_{xy}}{C_x}$$

where $C_x = \sum_y C_{xy}$. We will refer to the graph with conductance C as (G, C) . Informally, one can think about a person performing a walk on the nodes of this graph transitioning from one node to one of its neighbors with probability proportional to their edge conductances. Because of the unpredictable nature of this walk, it is called *random walk* but some prefer *drunkard's walk* :) The nodes of this graph, thus, give states of this Markov chain. Based on the definitions so far, Figure 1 gives three different ways of viewing random walks on graphs.

One of the best parts of these Markov chains is that they are *reversible*: there exists a positive function $x \mapsto \pi(x)$ ³ on V such that

$$\pi(x)P_{xy} = \pi(y)P_{yx}$$

For such chains, π is *stationary*, that is, $\sum_x \pi(x)P_{xy} = \pi(y)$, or equivalently, $P^\top \pi = \pi$ because $\sum_x \pi(x)P_{xy} = \sum_x \pi(y)P_{yx} = \pi(y) \sum_x P_{yx} = \pi(y)$. In our case, $\pi(x)$ could be C_x (or any constant multiple of this function) since

$$C_x P_{xy} = C_x \frac{C_{xy}}{C_x} = C_{xy} = C_{yx} = C_y \frac{C_{yx}}{C_y} = C_y P_{yx}$$

²Each node has a bounded degree although the degrees may not be uniformly bounded.

³ $\pi(\cdot)$ does not have to be a probability measure and, in fact, for Markov chains with infinitely many states (corresponding to infinite graphs) $\pi(\cdot)$ will never be a probability measure in this thesis.

Lastly, let's imagine we are given $\pi(x)$ and P , and we want to construct an electric network that is associated with the same transition matrix P . We argue that assigning $\pi(x)P_{xy}$ as edge conductances should do it since

$$P_{xy_{new}} = \frac{\pi(x)P_{xy}}{\sum_y \pi(x)P_{xy}} = \frac{P_{xy}}{\sum_y P_{xy}} = P_{xy}$$

Remark 3.1. For simple random walks, following the literature, one can assign $C_{xy} = R_{xy} = 1$ for any edge (x, y) . Hence, C_x could correspond to the degree of node x .

Viewing a graph as an electric network is quite helpful because many concepts related to electric circuits correspond to the properties related to random walks on graphs! To exploit this connection, we will choose a source node s to inject current into the graph and a sink node t (or a set of nodes) to ground the electricity. This would introduce a flow into the graph in the form of current i . i_{xy} will define the amount of current passing through directed edge (x, y) ⁴ and i_x is the net signed current from node x to its adjacent nodes.⁵ Secondly, it will induce a voltage v_x for each node x . The sink node t will have a voltage of 0 as it is grounded. We will define the *effective resistance* of a graph with a given source s as

$$R_{\text{eff}} = \frac{v_s}{i_s}$$

The inverse of effective resistance is *effective conductance* denoted by C_{eff} . We will sometimes refer to the current as the *electric flow* as it is an example of a *flow*.

Definition 3.2. A *flow* θ from s to t (s can be a single node or a collection of nodes) is an assignment of numbers to *directed* pairs of nodes (x, y) such that

- (1) $\theta_{xy} = -\theta_{yx}$
- (2) $\theta_x = \sum_y \theta_{xy} = 0$ if $x \neq s, t$
- (3) $\theta_{xy} = 0$ if x and y are not adjacent.

For a directed edge σ , we will also use the notation $\theta(\sigma)$ to refer to the flow passing through the edge.

Flows persevere energy.

⁴For conductance, the order of the edge does not matter as $C_{xy} = C_{yx}$. For current, it matters as $i_{xy} = -i_{yx} \neq i_{xy}$

⁵Then, i_s is the total amount of current injected into the circuit.

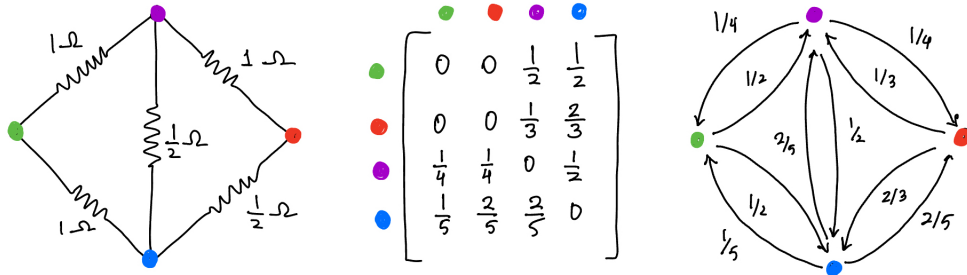


FIGURE 1. (left) an electric network with resistances (right) directed edges with transition probabilities based on conductances (middle) associated probability transition matrix

Theorem 3.3. (*Conservation of Energy*) Let f be any functions defined on the nodes of the graph and θ be a flow from some source s and sink t . Then,

$$(f(s) - f(t))\theta_s = \frac{1}{2} \sum_{x,y} (f(x) - f(y))\theta_{xy}$$

Proof.

$$\begin{aligned} \sum_{x,y} (f(x) - f(y))\theta_{xy} &= \sum_x (f(x) \sum_y \theta_{xy}) - \sum_y (f(y) \sum_x \theta_{xy}) \\ &= f(s) \sum_y \theta_{sy} + f(t) \sum_y \theta_{ty} - f(s) \sum_x \theta_{xs} - f(t) \sum_x \theta_{xt} \\ &= f(s)\theta_s + f(t)\theta_t - f(s)(-\theta_s) - f(t)(-\theta_t) \\ &= 2(f(s) - f(t))\theta_s. \end{aligned}$$

□

Definition 3.4. (Energy) If θ above is an electric flow i from s to t (so, $v_t = 0$) and $f(x)$ gives v_x . We get the equivalence

$$v_s i_s = \frac{1}{2} \sum_{x,y} (v_x - v_y) i_{xy}$$

Recall, $R_{\text{eff}} = \frac{v_s}{i_s}$. Then,

$$i_s^2 R_{\text{eff}} = \frac{1}{2} \sum_{x,y} (v_x - v_y) i_{xy} = \frac{1}{2} \sum_{x,y} i_{xy}^2 R_{xy} \quad (3.1)$$

which is *total energy dissipation* $\mathbf{E}$ in the circuit. When the current injected is unit, we get

$$R_{\text{eff}} = \frac{1}{2} \sum_{x,y} (v_x - v_y) i_{xy} \quad (3.2)$$

which will be very useful. This definition generalizes to any flow θ :

$$\theta_s^2 R_{\text{eff}} = \frac{1}{2} \sum_{x,y} \theta_{xy}^2 R_{xy}$$

gives the *energy* or *energy dissipation* for flow θ on graph with resistance R .

Electric flows are special flows in that they minimize energy dissipation!

Theorem 3.5. (*Thompson's Principle*) If i is the unit flow from s to t determined by Kirchhoff's Laws, i.e. it is an electric flow, then the energy dissipation $\frac{1}{2} \sum_{x,y} i_{xy}^2 R_{xy}$ minimizes the energy dissipation $\frac{1}{2} \sum_{x,y} j_{xy}^2 R_{xy}$ among all units flows j from a to b .

Proof. Let j be any unit flow from s to t . Let $k_{xy} = j_{xy} - i_{xy}$. Define $k_x = \sum_y k_{xy}$. Following Definition 3.2, it is not hard to show that k is a flow. By Theorem 3.3, we get

$$\sum_{x,y} (v_x - v_y) k_{xy} = 2(v_s - v_t)k_s = 2(v_s - v_t)(j_s - i_s) = 2(v_s - v_t)(1 - 1) = 0$$

Then,

$$\begin{aligned}
\sum_{x,y} j_{xy}^2 R_{xy} &= \sum_{x,y} (i_{xy} + k_{xy})^2 R_{xy} \\
&= \sum_{x,y} i_{xy}^2 R_{xy} + 2 \sum_{x,y} i_{xy} R_{xy} k_{xy} + \sum_{x,y} k_{xy}^2 R_{xy} \\
&= \sum_{x,y} i_{xy}^2 R_{xy} + 2 \sum_{x,y} (v_x - v_y) k_{xy} + \sum_{x,y} k_{xy}^2 R_{xy} \\
&= \sum_{x,y} i_{xy}^2 R_{xy} + \sum_{x,y} k_{xy}^2 R_{xy} \geq \sum_{x,y} k_{xy}^2 R_{xy}
\end{aligned}$$

□

Electric flows on graphs are fascinating because of probabilistic interpretations of effective resistance, current, and voltages which we will outline now.

- **Voltage:** When a unit voltage is applied between s and t so that $v_s = 1$ and $v_t = 0$, the voltage v_x at any node x is the probability that a walker starting from x will hit s before reaching t (also called the “hitting probability”).
- **Current:** When a unit current flows into s and out of t , the current i_{xy} is equal to the expected net⁶ number of times that a walker, starting at s and walking until she reaches t , will move along the edge directed (x, y) .

Related to but different from hitting probability, we have the probability of escape:

Definition 3.6. We define p_{esc} to be the probability that a walker starting from s reaches t before returning to s .

This should not be confused with hitting probability associated with v_s . v_s is always 1 since the random walk starts from s . p_{esc} deals with the second time the random walk hits s . It turns out that this probability is also connected to effective resistance or conductance in the following way:

Theorem 3.7.

$$p_{\text{esc}} = \frac{C_{\text{eff}}}{C_s} = \frac{1}{C_s R_{\text{eff}}}$$

where s is the source of the electric current as well as the starting point of the random walk.

Proof. Say we injected electric flow into the circuit from s such that $v_s = 1$, then

$$\frac{1}{R_{\text{eff}}} = \frac{v_s}{R_{\text{eff}}} = i_s$$

and

$$i_s = \sum_y (v_s - v_y) \frac{1}{R_{sy}} = \sum_y (v_s - v_y) \frac{C_{sy}}{C_s} C_s = C_s \left(1 - \sum_y P_{sy} v_y \right) = C_s p_{\text{esc}}$$

where we used the probabilistic interpretation of v_x (holds when $v_s = 1$). □

Remark 3.8. The nice part of this conclusion is that now our formula for p_{esc} is independent of how much current we inject into the graph and only depends on pre-assigned conductances (one can also think of them as weights) of the graph.

⁶If the walker moves along the other direction, we subtract the count.

We will skip the proofs for probabilistic interpretations of current and voltage since Peter G. Doyle and J. Laurie Snell have done a great job in explaining these concepts in their notes titled “Random walks and electric networks” [3]. The reader is strongly encouraged to check it out!

We will, however, dive deeper into what p_{esc} is when the underlying graph is infinite. In this case, p_{esc} will correspond to whether the random walker will ever return to its starting position or wander off far away which we call ∞ in a rather hand-wavy way. Hence, the probability we are interested is p_{esc} in the limit. Let’s make this more formal.

3.1. Infinite Graphs.

Definition 3.9. Let $\{X_n^s\}$ be a random walk on a graph started at node s . The random walk is defined as:

- **Recurrent:** if $\mathbb{P}(T_s < \infty) = 1$
- **Transient:** if $\mathbb{P}(T_s < \infty) < 1$

where T_s is the return time to s .

We will call the problem of determining the recurrence or transience of a random walk the **type problem**. For finite connected graphs, all random walks are recurrent since any path associated with the graph has a positive probability of occurring. So, if we let the random walker walk forever, at some point, she is guaranteed to come back to her starting location! But, this problem becomes very interesting and fun in the context of infinite graphs, and effective resistance turns out to be helpful again.

Take an infinite graph G . Let $B_r^{(G)}(s)$ be the subgraph of G induced by all the nodes of G that are reachable by traversing through at most r edges from node s . As before, the starting point of the random walk will be the node s where we will inject electricity. Unlike before, though, we will not ground any nodes in this graph. Alternatively, one can also imagine that we are grounding the set of nodes at the boundary of $\partial B_r^{(G)}(s)$ but, in the limit as $r \rightarrow \infty$, no node would be grounded.

Definition 3.10. Given an infinite graph G with a chosen source now s , we will define the effective resistance of this graph as

$$R_{\text{eff}} = \lim_{r \rightarrow \infty} R_{\text{eff}}^r$$

where R_{eff}^r is the effective resistance of $B_r^{(G)}(s)$ when s is the source and $\partial B_r^{(G)}(s)$ are the sinks. Then,

$$\mathbb{P}(T_s = \infty) = \frac{1}{C_s R_{\text{eff}}}$$

In particular, if $\{X_n^s\}$ is a simple random walk on G , then

$$\mathbb{P}(T_s = \infty) = \frac{1}{\deg(s) R_{\text{eff}}}$$

Remark 3.11. The most important point to notice in this section is that, based on the about formulation, the random walk on a graph is transient if R_{eff} is finite and recurrent if R_{eff} is infinite!

With this terminology, we can now introduce tools that will help us analyze random walks from the electric perspective.

Theorem 3.12. *The effective resistance R_{eff} from s to ∞ is bounded above by the energy dissipation of any flow from s to ∞ .*

Proof. Take a flow θ from s to ∞ with energy dissipation

$$E = \frac{1}{2} \sum_{x,y} \theta_{xy}^2 R_{xy}$$

Consider the subgraph induced by edges in $B_r^G(s)$. Now, let's induce 1 unit electric flow i from source s to sinks $\partial B_r^G(s)$ in $B_r^G(s)$ (so, $s_i = 1$). Since electric flow minimizes the energy dissipation among all other flows including θ , we have

$$R_{\text{eff}}^r = i_s^2 R_{\text{eff}}^r = \frac{1}{2} \sum_{(x,y) \in B_r^G(s)} i_{xy}^2 R_{xy} \leq \frac{1}{2} \sum_{(x,y) \in B_r^G(s)} \theta_{xy}^2 R_{xy} \leq \frac{1}{2} \sum_{(x,y) \in G} \theta_{xy}^2 R_{xy} = E$$

Taking $r \rightarrow \infty$, we prove the result. $\square$

This means that if we prove any unit flow (not necessarily electric flow!) has finite energy dissipation on a graph, then we know that effective resistance is finite. Thus, by Remark 3.11, the walk on the graph is transient!

Theorem 3.13. (*Rayleigh's Monotonicity Law*) *If the resistances of a circuit are increased, the effective resistance R_{eff} between any two points can only increase. If they are decreased, it can only decrease.*

Proof. Let i be the unit current flow from s to t with resistances R_{xy} . Similarly, let j be the unit current flow from s to t with resistances $\bar{R}_{xy}$. Now, assume $\bar{R}_{xy} > R_{xy}$ for all (x, y) . Then, by equation 3.1,

$$i_s^2 \bar{R}_{\text{eff}} = \frac{1}{2} \sum_{x,y} j_{xy}^2 \bar{R}_{xy} \geq \frac{1}{2} \sum_{x,y} j_{xy}^2 R_{xy} \geq \frac{1}{2} \sum_{x,y} i_{xy}^2 R_{xy} = i_s^2 R_{\text{eff}}$$

where the second last inequality holds because i minimizes energy dissipation with respect to resistance R_{xy} . Since both i and j are unit flows, i_s^2 and j_s^2 cancel out, giving the result. $\square$

The Monotonicity Law coupled with Remark 3.11 is very powerful. In particular, if we know that a random walk on a graph is recurrent, then $R_{\text{eff}} = \infty$. Deleting edges from the graph corresponds to making the corresponding edges have infinite resistance. Hence, if we delete any edge, the effective resistance can only increase and the random walk will be guaranteed to be recurrent since $R_{\text{eff}} = \infty$ will still hold. Likewise, if the random walk is transient and we add edges to the graph (corresponding to making edges of infinite resistance finite), it will still be transient. This strategy has a name: **Rayleigh's short-cut method**.

Theorem 3.14. *If for every edge (x, y) of G , we have $0 < u \leq C_{xy} \leq v < \infty$ for some constants u and v , then the random walk on (G, C) has the same type as a simple random walk on G .*

Proof. Let $U_{xy} = u$ and $V_{xy} = v$ for all $(x, y) \in G$. Notice that the random walks on (G, U) and (G, V) are both *simple* random walks, so they are the same *type*! Now consider the random walk on (G, C) where $U_{xy} \leq C_{xy} \leq V_{xy}$ and the resistance $R_{xy} = \frac{1}{C_{xy}}$ for each edge (x, y) . Let R_{eff}^V , R_{eff}^C and R_{eff}^U be the effective resistance of graph G with conductance V, C and U respectively. We know $\frac{1}{V_{xy}} \leq R_{xy} \leq \frac{1}{U_{xy}}$. Then, by the Monotonicity Law, we get

$$R_{\text{eff}}^V \leq R_{\text{eff}}^C \leq R_{\text{eff}}^U$$

Hence, if the random walk on (G, V) is recurrent, i.e. $R_{\text{eff}}^V = \infty$, then the random walk on (G, C) is recurrent too, i.e. $R_{\text{eff}}^C = \infty$. Likewise, if the random walk on (G, U) is

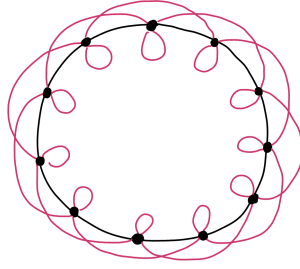


FIGURE 2. 2-fuzz of 11-cycle

transient, i.e. $R_{\text{eff}}^U < \infty$, then the random walk on (G, C) is also transient, i.e. $R_{\text{eff}}^C < \infty$. Thus, the random walk on (G, C) has the same type as a simple random walk. $\square$

An important consequence of this theory is Polya's theorem for random walks on lattices of arbitrary dimensions.

Theorem 3.15. (*Polya's Theorem*) *Simple random walk on a d -dimensional lattice is recurrent for $d = 1, 2$ and transient for $d > 2$.*

The key tool in this proof is Rayleigh's short-cut method and embedding trees into lattices. We will not give proof of this, but curious readers can check the beautiful explanation of Doyle & Snell for the proof [3]. On the other hand, we will dive deeper into trees in the next chapter and will now prove another fun theorem that uses the same key ideas in its proof and, as a result, shows a way to identify the type of random walk on a graph based on its drawing. This is very fascinating since it connects these concepts with the visual properties of a graph.

3.2. Drawings of Graphs.

Theorem 3.16. *If G can be drawn on paper (or $\mathbb{R}^2$) in such a way that for some $r < \infty$, $d > 0$*

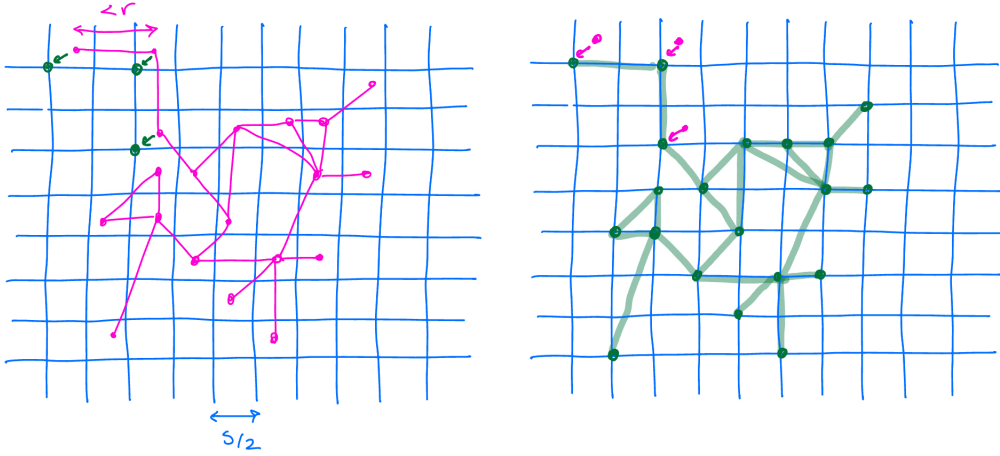
- *The length of each edge is less than or equal to r*
- *The distance between any two nodes is more than d*

then we say that graph G can be drawn "nicely," and when this is the case, random walk on G is recurrent.

Note: We will also assume that G is connected as always and has a degree bounded by some Δ .

Proof. Let's first define the k -fuzz of a graph G to be the graph G_k obtained from G by adding an edge (x, y) if it is possible to go from x to y in at most k steps (double edges and loops are included). See Figure 2 for the 2-fuzz of a cycle. The proof of this theorem involves proving several lemmas:

- (1) Simple random walk on 2-d lattice $\mathbb{Z}^2$ is recurrent (follows from Polya's Theorem).
- (2) If a graph can be drawn "nicely" on paper, then it can be embedded in a k -fuzz of lattice $\mathbb{Z}^2$ for some k .
- (3) Simple random walks on graphs F and G are of the same type (transient or recurrent) if each graph can be embedded in a k -fuzz of the other graph.

FIGURE 3. Embedding a graph in $\mathbb{Z}^2$

Lemma 3.17. *If a graph can be drawn “nicely” on paper for some r and d , then it can be embedded in a k -fuzz of lattice $\mathbb{Z}^2$, denoted by $\mathbb{Z}_k^2$, for some k .⁷*

Proof. Say we have a graph G that is drawn “nicely” (corresponding to the pink graph in Figure 3). Consider $\mathbb{Z}^2$ such that each edge has length $d/2$ (corresponding to the blue lattice in Figure 3). Each square of $\mathbb{Z}^2$ contains at most a single node of G . Hence, one can move each node v to the bottom-left corner of the square it is part of in $\mathbb{Z}^2$ (illustrated by the green graph in Figure 3). Since any two nodes x and y are at most r apart from each other, after these two nodes are translated to bottom-left corners to be $\hat{x}$ and $\hat{y}$, the length of the shortest path between $\hat{x}$ and $\hat{y}$ would be at most $r + d$. Hence, the edges between $\hat{x}$ and $\hat{y}$ (corresponding to the green edges in Figure 3) would exist in the k -fuzz of $\mathbb{Z}^2$ for $k = r + d$. $\square$

Lemma 3.18. *Simple random walks on graphs F and G are of the same type if each graph can be embedded in a k -fuzz of the other graph.*

Proof. Without loss of generality, say G can be embedded in F_k . Let P be the probability transition matrix of a simple random walk on F . Define a new random walk on the vertices of the graph with probability transition matrix $\bar{P} = (P + P^2 + \dots + P^k)/k$. $\bar{P}$ is relevant because it turns out the simple random walk on F_k and the random walk on F with probability transition matrix $\bar{P}$ are of the same type! (For now, we will assume this. The proof is given in Lemma 3.19.)

Let X_n^s be the random node a random walker starting from node s and walking according to probability transition matrix P is on at time n . Let U_n be an i.i.d random variable selected from $\{1, 2, \dots, k\}$ uniformly at random. Then, define another random walk $\{Y_n^s\}_{n=0}^\infty$ starting from s based on the following recurrence:

$$Y_{n+1}^o = Y_n^o + X_{n+U_n}^o = X_{\sum_{i=1}^n U_i}^o$$

Notice that Y_n moves according to the probability transition matrix $\bar{P}$! Hence, any random walk $\{Y_n\}_{n=0}^\infty$ following $\bar{P}$ could be interpreted as a sub-sequence of $\{X_n\}_{n=0}^\infty$ defined according to the above recurrence. Equivalently, one can think of $\{X_n\}_{n=0}^\infty$ as a random walk constructed by picking a path of length at most k from Y_n to Y_{n+1}

⁷Specifically, for any $k \geq r + d$.

uniformly at random according to P for each $n = 0, 1, 2, \dots$. According to this construction, $\{X_n^s\}_{n=0}^\infty$ will visit the starting node s (or any node) at least once for every time $\{Y_n\}_{n=0}^\infty$ visits that node. Hence, if a random walk according to $\bar{P}$ recurrent so is the one according to P .⁸

Now, let's assume $\bar{P}$ gives a transient random walk. Then, with positive probability, $\{Y_n\}_{n=1}^\infty$ would **not** visit the starting node. Let S be the set of nodes in F that are reachable from the starting node in k steps. Then, for any $x \in S$, $\bar{P}_{x,s} > 0$. If $\{Y_n\}_{n=1}^\infty$ visits nodes in S infinitely often, then the probability that it won't visit the starting node s will tend to 0. Since we know that this probability is lower bounded by a positive number, $\bar{P}^*$ moves out of S indefinitely after some, possibly large, amount of *finite* steps. If $\{Y_n\}_{n=1}^\infty$ never is part of S after some time, $\{X_n^s\}_{n=1}^\infty$ can never visit s after some time either. Therefore, if a random walk according to $\bar{P}$ transient so is the one according to P .

Thus, the random walks in F and F_k are of the same type. By the same logic, the random walks in G and G_k are of the same type. Lastly, we know that F can be embedded in G_k for some k . Since deleting edges (corresponding to increasing the edge resistances to infinity) would only increase effective resistance, according to Rayleigh's Monotonicity Law, the effective resistance of F would be larger than G_k (independent of where the source of the electricity is). Then, if G_k or G is recurrent, then F is recurrent too! By the same logic, if F_k or F is recurrent, then G is recurrent too. Therefore, the random walks in F and G are of the same type. $\square$

Lemma 3.19. *The following two random walks have the same type:*

- The simple random walk on F_k (Say this random walk has probability transition matrix P)
- The random walk on F_k with probability transition matrix

$$\bar{P} = (P + P^2 + \dots + P^k)/k$$

Let P^s be the probability transition matrix for the simple random walk on F_k . First, it is important to clarify that P^s for the simple random walk on F_k is not $\bar{P}$ through a counterexample. Let graph F be a path of 3 nodes and $k = 2$. Then, P^s is the simple random walk on F_2 .⁹

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix} P^2 = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} P^s = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

$$\bar{P} = \frac{P + P^2}{2} = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix} \neq P^s$$

We can start the proof of Lemma 3.19 now.

Proof. By assumption, F_k is a connected graph with a bounded degree. Then, the probability transition matrix P for the original graph is *reversible* as we discussed at the beginning of this chapter, meaning that there exists a function π , namely π such that $\pi(x)$ corresponds to the degree of node x , that satisfies $P^\top \pi = \pi$. By definition,

⁸Formally, we know that the probability measure of random walks according to $\bar{P}$ that return back to the starting node is 1. Hence, the probability measure of instances of P that return back to the starting node should be at least 1 too.

⁹ $\bar{P}$ is more sensitive to the original structure of the graph whereas P^s eliminates the nuances between nodes that are k=2 diameter of each other.

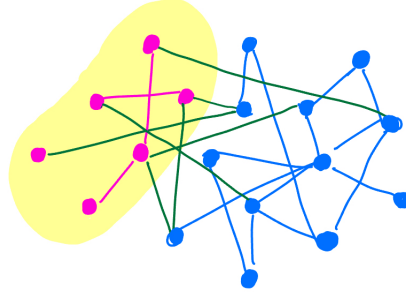


FIGURE 4. a graph with high conductance or expansion

$P^\top \pi = (P^\top \pi + P^{\top 2} \pi + \dots + P^{\top k} \pi)/k = (\pi + \pi + \dots + \pi)/k = \pi$ holds, so $\bar{P}$ is also reversible. Notice that $\bar{P}$ corresponds to the random walk on G_k such that the conductance $\bar{C}_{xy}$ between node x and y is $\pi(x)\bar{P}_{xy}$ as we discussed at the beginning of this chapter. Moreover, $\bar{P}_{xy} > 0$ if and only if there is a path from x to y of length at most k . Then,

$$1 \geq \bar{P}_{xy} \geq \frac{1}{k} \left(\frac{1}{\Delta} \right)^k > 0$$

where Δ is the upper bound on the degree. Thus,

$$\Delta \geq w_x = w_x \bar{P}_{xy} \geq \frac{1}{\Delta} \left(\frac{1}{\Delta} \right)^k > 0$$

Lastly, by Theorem 3.14, the simple random walk on G_k denoted by P^s and $\bar{P}$ are of the same type. $\square$

Now, let's put the pieces together to prove the original theorem! One can embed $\mathbb{Z}_k^2$ in the k -fuzz of $\mathbb{Z}^2$ by definition. The opposite embedding is also possible since $\mathbb{Z}^2$ is just a subgraph of $\mathbb{Z}_k^2$ for any k . Then, by Lemma (1) and Lemma 3.19, the simple random walk on $\mathbb{Z}_k^2$ is recurrent. Lastly, we know that G can be embedded in $\mathbb{Z}_k^2$ for some k . As before, by Rayleigh's Monotonicity Law, the effective resistance of G would be larger than $\mathbb{Z}_k^2$. Thus, the simple random walk on G would be recurrent too. $\square$

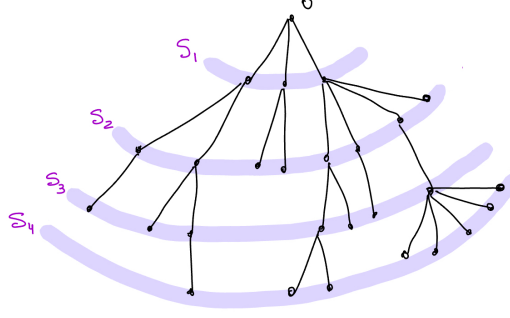
A “nicely” drawn graph would have its vertices spread out and not clumped together while having short edges at the same time. These are the characteristics of weakly connected graphs such as lattices.¹⁰ One way people characterize how connected finite graphs are is through their conductance.

Definition 3.20. Conductance Φ_G of a finite graph of a graph G with nodes V measures how “well-knit” the graph is and is defined as

$$\min_{S \subseteq V} \frac{\sum_{u \in S, v \in \bar{S}} \mathbb{1}_{(u,v) \in E(V)}}{\min(E(S), E(\bar{S}))}$$

where $\bar{S}$ is the complement of S and $E(F)$ corresponds to the edges in the induced subgraph of G by nodes in F . Although related concepts, one should **not** confuse Φ_G with the edge conductances or effective conductance we mentioned earlier in the context of electric flows.

¹⁰This is not an if and only if argument since 3D lattice embeds poorly in $\mathbb{R}^2$ but it is still weakly connected.

FIGURE 5. S_n : sphere of radius $n = 1, 2, 3, 4$

A well-knit graph visually would look more like a random graph as opposed to a lattice that is highly structured. Randomness helps because it ensures edges are not correlated with each other. For finite graphs, well-connected graphs are characterized by a low second eigenvalue of the Laplacian.¹¹ This is the consequence of the famous Cheeger's inequality connecting the expansion properties of a graph to its spectrum.

Theorem 3.21.

$$\frac{\lambda_2}{2} \leq \Phi_G \leq \sqrt{2\lambda_2}$$

where λ_2 is the second smallest eigenvalue of the Laplacian matrix and Φ_G is the conductance of G .

Expanders are one such finite highly connected graphs. They are special because they are sparse too, ensuring connectivity with few edges. They have many theoretical but also real-life applications [5]!

4. BRANCHING AND PERCOLATION

Now, we turn our attention to random walks on trees. Let's define some useful concepts to analyze them.

4.1. Branching Number.

Definition 4.1. A *cutset* Π of a tree T is a finite set of vertices excluding the root o such that, if removed from the tree, the connected component including the root, T_o , would be finite and such that no vertex in Π is part of the shortest path from the root to another vertex in Π . For example, for any graph, the sphere of radius r , S_r , gives a cutset (see Figure 5).

The *branching number* of a tree T , denoted by brT , is defined by

$$br(T) := \inf\{\lambda > 0 : \inf_{\Pi} \sum_{\sigma \in \Pi} \lambda^{-|\sigma|} = 0\} \quad (4.1)$$

where Π is any cutset of T and $|\sigma|$ corresponds to the length of the shortest path from the root o to vertex $\sigma \in T$. (Except for the root, each σ could be associated in a

¹¹The Laplacian L is defined as $D - DP = D - A$ where D is the diagonal matrix such that D_{ii} gives the degree of node i and P is the probability transition matrix of G as before, so $A = DP$ gives the adjacency matrix of a graph assuming that each has the same weight (or resistance/conductance).

Theorem 4.3. (*Lyons, 1990*) *Let T be a locally finite infinite tree. When $\lambda > \text{br}T$, the walk associated with λ is recurrent, and when $\lambda < \text{br}T$, the walk associated with λ is transient.*

Proof. Let s be the root of the tree. Say the walk associated with λ is transient. Then, R_{eff} is finite or, by equation 3.2, equivalently, there exists a unit electric flow i , i.e. $i_s = 1$, of finite energy:

$$E = i_s^2 R_{\text{eff}} = R_{\text{eff}} = \frac{1}{2} \sum_{x,y} i_{xy}^2 R_{xy} = \sum_{\sigma} i_{\sigma}^2 R_{\sigma} = \sum_{\sigma} i_{\sigma}^2 \lambda^{|\sigma|} < \infty$$

The $1/2$ factor disappeared after the fourth equality above since while pairs (x, y) for current i were ordered, we will assume that any edge σ is unordered: i_{σ} is considered to be the value of the current going away from the root through edge σ which cannot be negative as the current is injected into the root.

Now, notice that for any cutset Π , $1 \leq \sum_{\sigma \in \Pi} i_{\sigma}$ since all flow should pass through the cutset. Then, by Cauchy-Schwartz inequality,

$$1 \leq \left(\sum_{\sigma \in \Pi} i_{\sigma} \right)^2 \leq \left(\sum_{\sigma \in \Pi} \lambda^{-|\sigma|} \right) \left(\sum_{\sigma \in \Pi} i_{\sigma}^2 \lambda^{|\sigma|} \right)$$

This equation is true for all cutsets, specifically for S_n .¹² Notice that one can write

$$\sum_{\sigma} i_{\sigma}^2 \lambda^{|\sigma|} = \sum_n \sum_{\sigma \in S_n} i_{\sigma}^2 \lambda^{|\sigma|} < \infty$$

Then, as $n \rightarrow \infty$, we get $\sum_{\sigma \in S_n} i_{\sigma}^2 \lambda^{|\sigma|} \rightarrow 0$. Then, it must be that $\sum_{\sigma \in S_n} \lambda^{-|\sigma|} \rightarrow \infty$ as $n \rightarrow \infty$. Thus, we get that $\lambda \leq \text{br}(T)$ by definition of branching number. By contrapositive argument, $\lambda > \text{br}(T)$ implies that the walk associated with λ is recurrent.

Now, assume $\lambda < \text{br}(T)$ and choose $\gamma \in (\lambda, \text{br}(T))$. Next, set $w_n = \left(\frac{\lambda}{\gamma}\right)^n$. Since $w_0 < 1$, we get that $\sum_n w_n < \infty$ by sum of geometric series. We also know that

$$\inf_{\Pi} \sum_{\sigma \in \Pi} w_{|\sigma|} \lambda^{-|\sigma|} = \inf_{\Pi} \sum_{\sigma \in \Pi} \left(\frac{\lambda}{\gamma}\right)^{|\sigma|} \lambda^{-|\sigma|} = \inf_{\Pi} \sum_{\sigma \in \Pi} \gamma^{-|\sigma|} = L > 0$$

by our assumption that $\gamma < \text{br}(T)$ and definition of branching number.

Consider the capacity of each edge $\sigma \in \Pi_n$ to be $\gamma^{-|\sigma|} = w_{|\sigma|} \lambda^{-|\sigma|}$. By the Max-Flow Min-Cut theorem, we know that there exists a flow θ from the root of this tree to its leaves whose strength is L (importantly positive) and obeys $\theta(\sigma) \leq \gamma^{-|\sigma|} = w_{|\sigma|} \lambda^{-|\sigma|}$. The total energy dissipation of flow θ on the graph with resistances $\lambda^{|\sigma|}$ for each edge σ would satisfy:¹³

$$E = \sum_{\sigma \in T} \theta(\sigma)^2 \lambda^{|\sigma|} = \sum_n \sum_{\sigma \in S_n} \theta(\sigma)^2 \lambda^{|\sigma|} \leq \sum_n \sum_{\sigma \in S_n} \theta(\sigma) w_{|\sigma|} \lambda^{-|\sigma|} \lambda^{|\sigma|} = \sum_n w_n \sum_{\sigma \in S_n} \theta(\sigma)$$

Finally,

$$E \leq \sum_n w_n \sum_{\sigma \in S_n} \theta(\sigma) = L \sum_n w_n < \infty$$

¹²Recall that S_n is the cutset corresponding to the sphere of radius n . In other words, S_n is the set of nodes such that the length of the shortest path between any node $v \in S_n$ and the root is n . Importantly, S_n for $n = 0, 1, 2, \dots$ partitions the nodes of any tree to disjoint sets since trees do not have any cycles.

¹³Again, $1/2$ disappears since each edge σ is undirected.

as $\sum_{\sigma \in S_n} \theta(\sigma) = L < \infty$.¹⁴ By Theorem 3.12, the energy of this flow gives an upper bound for the effective resistance of T with resistances satisfying $\lambda^{|\sigma|}$ for $\sigma \in T$. This implies that the effective resistance of T with the root as the source is finite. Hence, by Remark 3.11, the random walk on T starting from the root is transient. $\square$

4.2. Percolation. Next, we will prove that the branching number of a (possibly random) tree is the inverse of its critical percolation threshold. To do so, we first need to understand percolation. Given a graph G and probability p , the percolated graph G_p is the random graph obtained by including each edge of G with probability p independently from each other. One can view this as the thinning of edges of G with probability $1 - p$ as well.

Definition 4.4. (Erdős-Rényi) If the original graph G is a complete graph with n nodes, the corresponding percolated G_p is the famous **Erdős-Rényi** graph model denoted by $G(n, p)$.

As is the case for $G(n, p)$, many people are interested in the threshold for p for the emergence of certain structures in this graph. One such structure is the *infinite connected component* containing the root o of the graph, defined as $\mathcal{C}_o$. Such a component cannot occur if the original G is finite (e.g. Erdős-Rényi) but it is worth investigating if G is an infinite graph. Since this structure is widely studied, its threshold for p is referred to as the *critical percolation threshold* for a graph G and is defined as follows:

Definition 4.5.

$$p_c(G) = \sup\{p \in [0, 1] : \mathbb{P}_p[|\mathcal{C}_o| = \infty | G] = 0\}$$

Percolation theory was developed in the 1950s with the work of scientists like Simon Broadbent and John Hammersley to study the movement and filtering of fluids through porous materials [2]. It was first studied on lattices since lattices could model the porous structure of materials: each node could represent a potential pore and edges represent fluid pathways. Following them, we will also first explore the critical percolation threshold for lattices and then turn back to trees later on.¹⁵

4.3. Lattices. Let's start with the simple 1-dimensional lattice: $\mathbb{Z}$.

Lemma 4.6. $p_c(\mathbb{Z}) = 1$

Proof. Take any $p < 1$. $|\mathcal{C}_o| = \infty$ if and only if the root is connected to infinitely many nodes on its right, on its left, or both. By union bound, we get $\mathbb{P}_p[|\mathcal{C}_o| = \infty] \leq 2p^\infty \rightarrow 0$ since $p < 1$. Thus, $p_c(\mathbb{Z}) = 1$. $\square$

One may think that $p_c(\mathbb{G})$ is always 1 since one needs infinitely many edges to survive to have an infinite component. However, in certain infinite graphs, the number of potential infinite graphs containing the root is also infinite! Hence, we may not need p as large as 1 so that a single infinite component survives. To demonstrate this, let's move on to the 2-dimensional lattice $\mathbb{Z}^2$ and show that $p_c(\mathbb{Z}^2)$ is non-trivial ($p_c(G) \notin \{0, 1\}$). We will give rather loose lower and upper bounds on p_c .

¹⁴To see this, notice $L \leq \sum_{\sigma \in S_1} \gamma^{-1} = \deg(s) \frac{1}{\gamma} < \infty$ for positive γ and a locally finite graph.

¹⁵Note that since infinite lattices are completely symmetrical the choice of the root is arbitrary. For trees, the choice of the root is obvious.

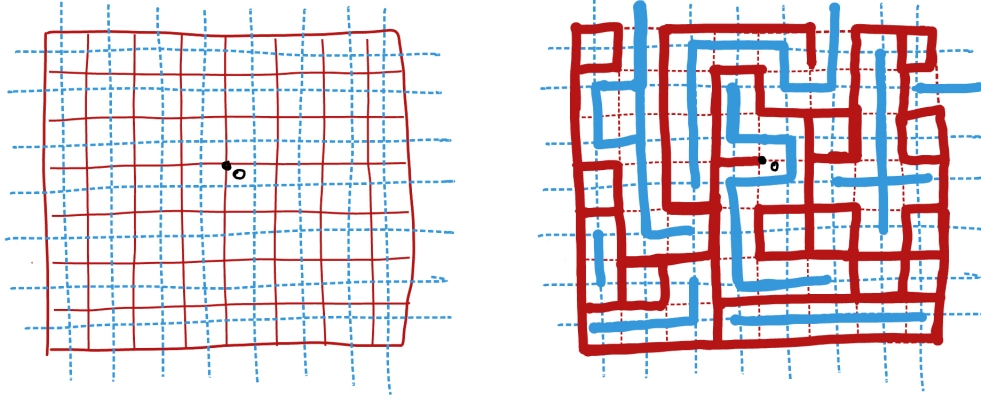


FIGURE 7. (left) Red graph is a section of $\mathbb{Z}^2$ and the blue graph is the corresponding section of its dual graph $(\mathbb{Z}^2)^*$ (right) thick red edges give the red graph post-percolation $\mathcal{C}_o$ where o is the root and the thick blue graph corresponds to its dual $(\mathcal{C}_o)^*$

Theorem 4.7. $1/3 \leq p_c(\mathbb{Z}^2) \leq 2/3$

Proof. Let's start with the lower bound: First notice $|\mathcal{C}_o| = \infty$ if and only if there is a self-avoiding path of ∞ -length. Hence, instead of $\mathbb{P}_p[|\mathcal{C}_o| = \infty | \mathbb{Z}^2]$ ¹⁶ we will consider the probability of the latter event. Now, the number of self-avoiding paths from o of length n is at most $4 \cdot 3^{n-1}$ as any such path cannot return the direction it came from. The probability of a specific path of length n surviving post-percolation is p^n . By union bound, the probability that there exists a self-avoiding path of length n is at most $4 \cdot 3^{n-1} p^n$. If $p < 1/3$, then $4 \cdot 3^{n-1} p^n \rightarrow 0$ as $n \rightarrow \infty$. As such, $\mathbb{P}_p[|\mathcal{C}_o| = \infty | \mathbb{Z}^2] \leq 0$. Then, $p_c(\mathbb{Z}^2) \geq 1/3$. For the upper bound, the key step is to consider the dual of $\mathbb{Z}^2$. The dual of a lattice is a lattice obtained by placing a node in the middle of each small square in the non-percolated original lattice and connecting it to create another $\mathbb{Z}^2$. Let's call this $(\mathbb{Z}^2)^*$. This is best understood visually, so please check out Figure 7. Crucially, observe that for each edge e in the non-percolated original lattice G , there is another edge e^* in the new lattice G' that intersects it. Post-percolation of $\mathbb{Z}^2$, we will say an edge e^* in $(\mathbb{Z}^2)^*$ survived if corresponding e in $\mathbb{Z}^2$ did not survive the percolation (see Figure 7 for the dual of percolated $\mathbb{Z}^2$). Then, the probability that an edge survives in $(\mathbb{Z}^2)^*$ is $1 - p$.

Now, observe that $|\mathcal{C}_o| < \infty$ in $\mathbb{Z}^2$ if and only if there exists a simple finite cycle in $(\mathbb{Z}^2)^*$ surrounding o consisting of all closed edges. As such, similar to before, we will calculate the probability of this event instead. Let A_n be the event that such a simple cycle exists in $(\mathbb{Z}^2)^*$ and has length n .

Let's first bound the number of such cycles around node o of length n . To do so, notice that any cycle in $(\mathbb{Z}^2)^*$ around node o has to cross from one of n nodes to the right of o in $(\mathbb{Z}^2)^*$ (see Figure 8 for illustration). Hence, the number of cycles of length n around o could be upper bounded by the number paths of length n from each of these

¹⁶Here, conditioning is unnecessary. We are doing so to avoid confusion when we introduce a new graph. Spoiler :)

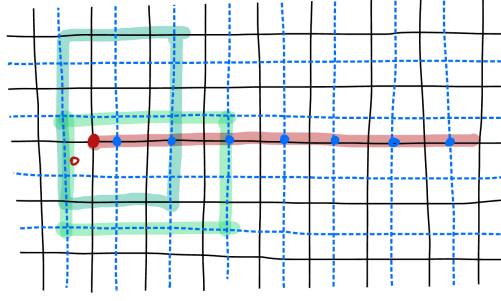


FIGURE 8. Cycles of length 8 (green shaded edges) cross one of 8 nodes to the right of node o (red shaded nodes) in $(\mathbb{Z}^2)^*$

n nodes, which is bounded by $4n(3^{n-1})$ as per the calculation before. This is a pretty loose bound but it works!

Now, assume that $p > 2/3$, meaning that $1 - p < 1/3$. Then,

$$4n(3^{n-1})(1-p)^n < \frac{4}{3}nx^{n-1}$$

for some $x < 1$. Then,

$$\sum_{n=4}^{\infty} \frac{4}{3}nx^{n-1} = \frac{4}{3} \left(4 \sum_{n=3}^{\infty} x^n + \sum_{n=4}^{\infty} \sum_{m=n}^{\infty} x^m \right) = \frac{4}{3} \left(4 \frac{x^3}{1-x} + \frac{x^4}{(1-x)^2} \right)$$

converges. Then, for some large enough N ,

$$\mathbb{P}_p[\cup_{n \geq N}^{\infty} A_n | (\mathbb{Z}^2)^*] \leq \sum_{n=N}^{\infty} \mathbb{P}_p[A_n | (\mathbb{Z}^2)^*] = \sum_{n=N}^{\infty} 4n(3^{n-1})(1-p)^n < \sum_{n \geq N}^{\infty} \frac{4}{3}nx^{n-1} < 1$$

This is the critical point of the proof where the duality is exploited! Take the event B_1 to be the event that all edges of $[-N, N] \times [-N, N]$ survived and B_2 to be the event that there is no completely survived simple cycle in the dual $(\mathbb{Z}^2)^*$ surrounding $[-N, N] \times [-N, N]$. Observe that $B_1 \cap B_2$ implies $|\mathcal{C}_o| = \infty$. To see this, assume B_1 and B_2 . Then, take any simple cycle in the dual $(\mathbb{Z}^2)^*$ surrounding $[-N, N] \times [-N, N]$. By B_2 , we know that some edge in $(\mathbb{Z}^2)^*$ is part of this cycle but did not survive. Then, by duality, some edge in $\mathbb{Z}^2$ connects the inner part of this cycle to the outer part of the cycle that has survived! (see Figure 9) Thus, there is always going to be an edge that will allow the connected component containing $[-N, N] \times [-N, N]$ in $\mathbb{Z}^2$ to extend more and more, meaning that $|\mathcal{C}_o| = \infty$. One can find all these edges that extend the original connected component greedily by choosing a cycle that *tightly* surrounds the connected component at each step.

Wrapping up the proof, since B_1 and B_2 deal with different edges, they are independent events. Then, we have $\mathbb{P}[\mathcal{C}_o = \infty] \geq \mathbb{P}_p[B_1 \cap B_2] = \mathbb{P}_p[B_1] \cdot \mathbb{P}_p[B_2]$. N is finite, so $\mathbb{P}_p[B_1] > 0$. Moreover, $\mathbb{P}_p[B_2] \geq 1 - \mathbb{P}_p[\cup_{n=4}^{N-1} A_n | (\mathbb{Z}^2)^*] > 0$. Then, $\mathbb{P}_p[B_1 \cap B_2] > 0$, suggesting

$$\mathbb{P}[\mathcal{C}_o = \infty] > 1 - \sum_{n \geq N}^{\infty} \frac{4}{3}x^{n-1} > 0$$

Thus, $p_c(\mathbb{Z}^2) \leq 2/3$. □

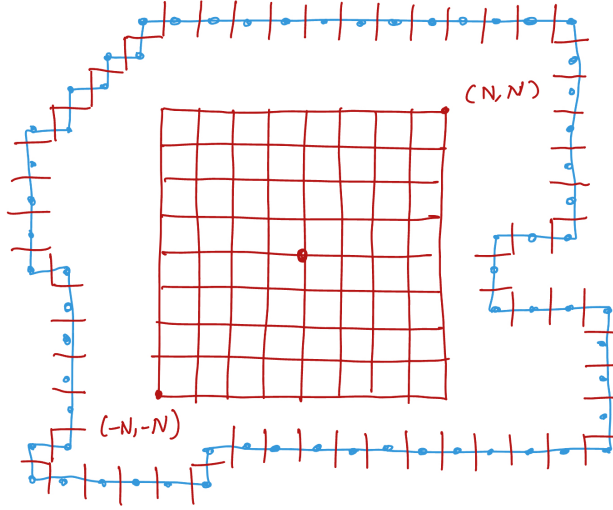


FIGURE 9. Red graph is a section of the original graph $\mathbb{Z}^2$. The blue graph is its dual $(\mathbb{Z}^2)^*$. For $N = 4$, the middle block gives $[-N, N] \times [-N, N]$. The blue cycle represents an arbitrarily picked cycle surrounding $[-N, N] \times [-N, N]$ in the dual. The core of the argument is B_2 implies one of the edges intersecting this blue cycle survives

Remark 4.8. The method of proof used here is often called a contour or Peierls argument. A more advanced version of this is used in proving the exact threshold for $\mathbb{Z}^2$ giving the following result.

Theorem 4.9. $p_c(\mathbb{Z}^2) = 1/2$.

Remark 4.10. Finding the percolation threshold in higher-dimensional lattices is challenging due to the exponential increase in possible configurations as dimensions rise (e.g. too many cycles), making an analysis more complex. Higher dimensions also lack the symmetries and simplifications present in 1D and 2D, not giving any clear arguments based on duals.

As we mentioned before, there are infinitely many potential infinite components in $\mathbb{Z}^2$ and $\mathbb{Z}^d$ for any $d > 2$ too. Above the percolation threshold, at least one of them will survive. Naturally, the next question is: how many infinite components are there above the percolation threshold?

Theorem 4.11. *The infinite component in the percolated d -dimensional lattice is unique if it exists.*

The proof of this theorem uses one of the most famous results of probability theory: Kolmogorov's Zero-One Law.

Theorem 4.12. *(Kolmogorov's Zero-One Law) Given a probability space $(\Omega, \mathcal{F}, \mu)$ and a sequence $\{A_n\}$ of independent events taken from this space, if we consider an event S from the tail σ -algebra $\mathcal{F}$ generated by the events of $\{A_n\}$, then $\mu(S)$ is either 0 or 1.*

Let's first understand what this theorem says: Ω is our outcome space, $\mathcal{F}$ refers to σ -algebra on the elements of Ω : the set of some subsets (of Ω in this case) that is

closed under union, complement, and intersection, and μ gives the probability of each subset of Ω . We will use this theorem without proof. The proof itself is based on the independence of events and is not that hard once the definition of the tail events is well understood.

Definition 4.13. The tail σ -algebra generated by the sequence $\{A_n\}$ is the intersection of smallest σ -algebras $\mathcal{F}_n$ containing $A_1, \dots, A_n$, and tail event is any event S that lies in this intersection:

$$S \in \cap_{i=1}^{\infty} \mathcal{F}_i$$

Intuitively, one can think of a tail event S as an event that is always independent of finitely many events from $\{A_n\}$.

We are ready to prove Theorem 4.11, which will give an example of a tail event.

Proof. Assume there exists an infinite component. First, we will argue that the event that there are exactly n number of infinite connected components in the percolated lattice is a tail event. Let's call this event S_n . To see this, let A_i be the event that the i^{th} edge is open. Note that a finite number of edges being closed or open does not affect whether there is an infinite component or not. Then, $S_n \in \cap_{i=1}^{\infty} \mathcal{F}_i$ is a tail event. Hence, $\mathbb{P}(S_n)$ is 0 or 1. Because S_k for $k \in \{1, 2, 3, \dots, \infty\}$ are disjoint events and $\mathbb{P}(\cup S_n) = \sum \mathbb{P}(S_n) = 1$ by assumption, we have $P(S_k) = 1$ exactly for one k . Surprisingly, it turns out that ruling out $k \in \{2, 3, \dots\}$ is easier than $k = \infty$. Here, we will only do the former. The reader is referred to [8] for $k = \infty$.

Without loss of generality, pick $k \in \{2, 3, \dots\}$. Assume for the sake of contradiction that $P(S_k) = 1$, so we have $P(S_1) = 0$. Let $F_M = \{\text{there are } k \text{ infinite clusters and each intersects } [-M, M]^d\}$. First, notice $F_1 \subseteq F_2 \subseteq \dots$ and $\cup F_i = S_k$. Therefore, $P(F_n) \rightarrow 1$. Then, there exists some N such that $P(F_N) > 0$ by the usual definition of limit.

Let $\bar{F}_N$ be the event that all k infinite clusters touch the boundary of, but not necessarily intersect, $[-N, N]^d$. This event is determined only by the events outside of $[-N, N]^d$. Let G be the event that all edges in $[-N, N]^d$ are open. Notice that this event is determined by the events inside of $[-N, N]^d$, so $\bar{F}_N$ and G are independent. Since N is finite, $P(G) > 0$. Secondly, observe $F_N \subseteq \bar{F}_N$, so $P(\bar{F}_N) \geq P(F_N) > 0$. Then,

$$P(S_1) \geq P(\bar{F}_N \cap G) = P(G) \cdot P(\bar{F}_N) > 0$$

This is a contradiction. Hence, $k = 1$ and the uniqueness of the infinite component follows. $\square$

Remark 4.14. This proof is interesting because it reflects a common technique we will touch on again later based on growing independent large (or infinite in this case) clusters and arguing that they have to intersect each other.

4.4. Trees. Finally, we are back to trees and branching numbers!

Theorem 4.15. (Lyons, 1990) For any tree T ,

$$p_c(T) = \frac{1}{brT}$$

Proof. Let o be the root as well as the source. Given a cutset Π and $\lambda > 0$, let

$$C_{\Pi}^{\lambda} = \sum_{\sigma \in \Pi} \lambda^{-|\sigma|}$$

(One can think of C_Π^λ as the capacity of cutset Π using the terminology of flows). Let Ω_p be the outcome space for the state of the tree T after percolation with probability p and let $T_o(\omega_p)$ be the connected component containing the root post-percolation. Also, define $\Pi(\omega_p) = T_o(\omega_p) \cap \Pi$. Then,

$$\mathbb{E}[\mathbb{Z}_{\Pi(\omega_p)}^1] = \sum_{\sigma \in \Pi} \mathbb{P}[\sigma \in T_o(\omega_p)] = \sum_{\sigma \in \Pi} p^{|\sigma|} = C_\Pi^{p^{-1}}$$

The second last inequality comes from the fact that for a vertex σ (or edge¹⁷) to be part of T_o after percolation, all of the edges in the shortest path from the root to this vertex σ have to survive.

Say $p^{-1} > brT$, then there exists a sequence Π_n such that $\mathbb{Z}_{\Pi_n}^{p^{-1}} \rightarrow 0$ as $|\Pi_n| \rightarrow \infty$ by definition of branching number. Then,

$$0 \leq \mathbb{E}[\liminf_{\substack{\Pi_n \\ |\Pi_n| \rightarrow \infty}} \mathbb{Z}_{\Pi_n i(\omega_p)}^1] \leq \liminf_{\substack{\Pi_n \\ |\Pi_n| \rightarrow \infty}} \mathbb{E}[\mathbb{Z}_{\Pi_n(\omega_p)}^1] = \liminf_{\substack{\Pi_n \\ |\Pi_n| \rightarrow \infty}} C_\Pi^{p^{-1}} \rightarrow 0$$

where the second inequality is due to Fatou's Lemma (see Appendix). This implies that

$$\liminf_{\substack{\Pi_n \\ |\Pi_n| \rightarrow \infty}} \mathbb{Z}_{\Pi_n(\omega_p)}^1 = \lim_{\substack{\Pi_n \\ |\Pi_n| \rightarrow \infty}} \sum_{\sigma \in \Pi_n} \mathbb{1}_{\{\sigma \in T_o(\omega_p)\}} = 0$$

almost surely, meaning that $|T_o(\omega_p)|$ is finite almost surely. Thus, $p_c(T) \leq \frac{1}{brT}$.

Now, assume the opposite: $p^{-1} < brT$. Now, let's inject one unit of flow into the tree from the root and define:¹⁸

$$X_n^\theta(\omega_p) = \sum_{\sigma \in S_n(\omega_p)} \theta(\sigma) p^{-|\sigma|}$$

where $\theta(\sigma)$ corresponds to the amount of flow that passes through edge σ before percolation.

Take $W = \{\omega_p \in \Omega_p | \sigma \in T_o(\omega_p)\}$ for $|\sigma| = 0, \dots, n$. Let $\mathcal{F}_n$ be the sigma-algebra generated by W .¹⁹ Note that X_n^θ is fully determined by $\mathcal{F}_n$ (i.e. it is measurable relative to $\mathcal{F}_n$) and $\mathbb{E}[X_n^\theta] \leq 1 < \infty$. Furthermore, we have

$$\mathbb{E}[X_{n+1}^\theta | \mathcal{F}_n] = \sum_{\sigma \in S_n} \sum_{\sigma \rightarrow \tau} p \theta(\tau) p^{-|\sigma|-1} = \sum_{\sigma \in S_n} \theta(\sigma) p^{-|\sigma|} = X_n^\theta$$

where $\sigma \rightarrow \tau$ indicates τ is children of σ in the branching process, the first equality is due to $\mathbb{E}[\mathbb{1}_{\tau \in S_{n+1}}] = p$ (the survival probability post-percolation) and the second equality is because the sum of flows that pass through child nodes gives the flow passing through the parent node pre-percolation. Thus, by definition, sequence $\{X_n^\theta\}$ is a martingale (see Appendix for all the following facts on Martingales).

Since $\{X_n^\theta\}$ is a nonnegative sequence by definition, by Martingale Convergence Theorem, it converges almost surely to some finite random variable X^θ . The conditions of the Martingale Convergence Theorem do not guarantee convergence of expectations. However, if the martingale is bounded in L^2 , the convergence in mean also holds. Then, let's show the bound in L^2 .

¹⁷Recall that every node except for the root has a unique edge that is associated with a tree, namely its edge connecting it to its parent vertex

¹⁸Recall S_n is the cutset corresponding to the sphere of radius n

¹⁹The set of sets that is closed under complement, countable unions, countable intersections, and contain all $w_p \in W$

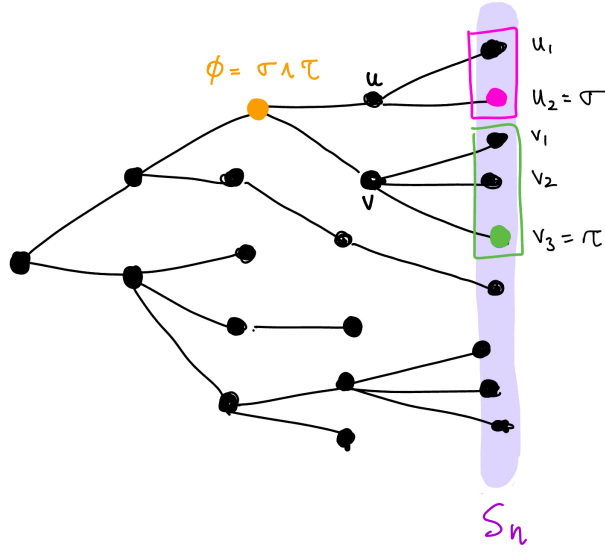


FIGURE 10. $\phi = \sigma \wedge \tau$, $\{\nu | \phi \rightarrow \nu\} = \{u, v\}$,
 $S_n^\phi = \{u_1, u_2, v_1, v_2, v_3\}$, $S_n^u = \{u_1, u_2\}$, $S_n^v = \{v_1, v_2, v_3\}$

To do so, for node σ and τ , define $\sigma \wedge \tau$ as the node furthest from the root both the shortest path from the root o to σ and the shortest path from the root o to τ contains. For example, in Figure 10, $\sigma \wedge \tau$ gives ϕ . Secondly, $\{\nu | \sigma \rightarrow \nu\}$ gives all the offspring of σ .²⁰ For example, in Figure 10, $\{\sigma | \phi \rightarrow \sigma\} = \{u, v\}$. Lastly, let S_n^ϕ be all the grandchildren of ϕ in S_n . In other words, S_n^ϕ is the set of all nodes $u \in S_n$ such that the shortest path from the root o to u contains ϕ . Then,

$$\begin{aligned}
 \mathbb{E}[(X_n^\theta)^2] &= \sum_{\sigma, \tau \in S_n} \theta(\sigma)\theta(\tau)p^{-2n}\mathbb{P}[\sigma, \tau \in T_o(\omega_p)] \\
 &= \sum_{\sigma, \tau \in S_n} \theta(\sigma)\theta(\tau)p^{-2n}(p^{|\sigma \wedge \tau|}p^{|\sigma| - |\sigma \wedge \tau|}p^{|\tau| - |\sigma \wedge \tau|}) \\
 &= \sum_{\sigma, \tau \in S_n} \theta(\sigma)\theta(\tau)p^{-|\sigma \wedge \tau|} = \sum_{|\phi| \leq n} p^{-|\phi|} \sum_{\substack{\sigma, \tau \in S_n \\ \sigma \wedge \tau = \phi}} \theta(\sigma)\theta(\tau) \\
 &= \sum_{|\phi| \leq n} p^{-|\phi|} \left(\left(\sum_{\sigma \in S_n^\phi} \theta(\sigma) \right)^2 - \sum_{\substack{\sigma, \tau \in S_n^\phi \\ \sigma \wedge \tau \neq \phi}} \theta(\sigma)\theta(\tau) \right)
 \end{aligned}$$

²⁰If σ is a leaf node, we have $\{\nu | \sigma \rightarrow \nu\} = \emptyset$.

Using $\theta(\phi) = \sum_{\sigma \in S_n^\phi} \theta(\sigma)$, we simplify the first term and get

$$= \sum_{|\phi| \leq n} p^{-|\phi|} \left[\theta(\phi)^2 - \sum_{\substack{\phi \rightarrow \nu \\ |\nu| \leq n}} \left(\sum_{\sigma \in S_n^\nu} \theta(\sigma) \right)^2 \right]$$

where the second term is due to the following reasoning: (1) $\sigma \wedge \tau = \phi$ implies $\sigma, \tau \in S_n^\phi$, and (2) for any two nodes $\sigma, \tau \in S_n^\phi$, $\sigma \wedge \tau \neq \phi$ if and only if $\sigma, \tau \in S_n^\nu$ for some offspring ν of ϕ or $\phi \rightarrow \nu$ (see Figure 10 for an example).²¹ Lastly, we have

$$= \sum_{|\phi| \leq n} p^{-|\phi|} \left[\theta(\phi)^2 - \sum_{\substack{\phi \rightarrow \nu \\ |\nu| \leq n}} \theta(\nu)^2 \right] \leq \sum_{|\phi| \leq n} p^{-|\phi|} \theta(\phi)^2$$

Then, by our assumption $p^{-1} < \text{br}(T)$ and Theorem 4.3, the walk associated with $\lambda = \frac{1}{p}$ is transient. This means that effective resistance or, equivalently, the *unit* electric flow associated with p^{-1} has finite energy. Let the unit flow θ we injected correspond to this electric flow since the choice of θ was arbitrary. Then, the sum $\sum_{|\phi| \leq n} p^{-|\phi|} \theta(\phi)^2$, corresponding the energy dissipation, is finite, implying $\mathbb{E}[(X_n^\theta)^2] < \infty$.

Then, $\mathbb{E}[X_0^\theta] = \mathbb{E}[X_1^\theta] = \dots = \mathbb{E}[X_n^\theta] \rightarrow \mathbb{E}[X^\theta]$ and, thus, $\mathbb{E}[X_0^\theta] = \mathbb{E}[X^\theta] = 1$ since the flow is unit. Then, $X^\theta > 0$ is positive with positive probability. This means that the tree does not go extinct with a positive probability. Therefore, $|T_o(\omega_p)|$ is infinite with positive probability. This implies $p_c(T) \geq \frac{1}{\text{br}(T)}$, concluding the proof. $\square$

This theorem holds for any tree, even the random ones. On this note, we introduce one of the most classic ways to generate random trees in the literature: *Galton-Watson branching process*!

Definition 4.16. The Galton-Watson branching process $\{Z_n\}_{n \geq 0}$ models the size of generations in a population. It starts with $Z_0 = 1$, a single node. Next, given that Z_n nodes are present in generation n , the number of offspring of nodes in generation n is given by i.i.d. random variables $X_{n,i}$ for $i = 1, \dots, Z_n$ and all n . Then, the size of the next generation is given by:

$$Z_{n+1} = \sum_{i=1}^{Z_n} X_{n,i}.$$

This is a single-type branching process since all nodes' offspring distributions are identical.

Define η as the probability of extinction, i.e., $\lim_{n \rightarrow \infty} \Pr(Z_n = 0)$. For the branching process to go extinct, each branch has to go extinct. Since each branch is independently distributed, η satisfies the closed form below:

$$\eta = G(\eta) = \mathbb{E}[\eta^X] = \sum_{k=0}^{\infty} p_k \eta^k$$

where $p_k = \Pr(X = k)$ and $G(0) = p_0$. When $p_0 = 0$, it is obvious that the branching process never goes extinct, so $\eta = 0$. Now, let's look into $p_0 > 0$. Since $G(1) = 1$, 1 is always a root of $G(\eta) = \eta$. Because $G''(\eta) = \sum_{k=0}^{\infty} p_k k(k-1) \eta^{k-2} > 0$ for $p_0 > 0$, G is strictly convex. Then, G is strictly increasing, so the final curvature of G at $\eta = 1$

determines whether $G(\eta) = \eta$ holds for any $\eta < 1$. Thus, when $G'(1) < 1$, the only root is $\eta = 1$. When $G'(1) > 1$, however, there exists one another root, and it is not that hard to show that η should correspond to the smallest non-negative root (refer to [7]). We will define the probability of non-extinction or survival as $\zeta = 1 - \eta$.

Lastly, notice $G'(1) = \sum_{k=1}^{\infty} p_k k \eta^{k-1} = \sum_{k=1}^{\infty} p_k k = \mathbb{E}[X]$, giving the next corollary.

Corollary 4.16.1. *If the mean degree of a tree generated by the Galton-Watson process is less than 1, then this tree is finite (i.e. $Z_t = 0$ for large enough t) almost surely.*

Definition 4.17. One can also have a similar branching process that generates a random tree that allows for nodes with k different offspring distributions as follows: Let $Z_n^{(j)}$ denote the number of entities of type j in the n -th generation. Then, $Z_n = \sum_{j=1}^k Z_n^{(j)}$ is the number of nodes in generation n as before. The dynamics of the process is described by:

$$Z_{n+1}^{(j)} = \sum_{i=1}^{Z_n^{(1)}} X_{n,i}^{(1,j)} + \sum_{i=1}^{Z_n^{(2)}} X_{n,i}^{(2,j)} + \cdots + \sum_{i=1}^{Z_n^{(k)}} X_{n,i}^{(k,j)}$$

where $X_{n,i}^{(l,j)}$ is a random variable representing the number of type j offspring of a type l node in n -th generation. In this model, for a fixed (k, j) , $X_{n,i}^{(k,j)}$ is independent and identically distributed for all n and i . Since different node types have different offspring distributions, this model is called a multi-type branching process.

Theorem 4.18. (Lyons, 1990) *Let the probability of k offspring be $0 \leq p_k < \infty$ in a Galton-Watson branching process. Then, if $m = \sum k p_k > 1$, the branching number of the tree is m a.s. (almost surely) conditioned on the branching process surviving.*

Proof. Let T be a random tree sampled from the Galton-Watson distribution μ_1 with mean offspring m . Now let's percolate on T with probability p . Notice that $T_o(\omega_p)$, the connected component containing the root post-percolation, is going to follow another Galton-Watson distribution μ_2 with mean offspring mp .

By Corollary 4.16.1, we know that when $mp < 1$, we have

$$\mathbb{P}_{T \sim \mu_1}(|T_o(\omega_p)| < \infty) = \mathbb{P}_{T \sim \mu_2}(|T| < \infty) = 1$$

Then, critical percolation probability $p_c(T) \geq \frac{1}{m}$. By Theorem 2.8, we get $br(T) \leq m$ a.s. Since the result holds almost surely, conditioned on the survival of T , $br(T) \leq m$ a.s. as well.

Now, we will prove that, conditioned on survival of the branching process, $br(T) \geq m$ a.s. Let S be the event that the branching process survived. Define

$$\lambda^* = \sup\{x | x \leq m, \mathbb{P}[br(T) > x] > 0\}$$

For any $\lambda > \lambda^*$, we know that $\mathbb{P}[br(T) > \lambda] = 0$ by definition. Conditioned on survival of T , $\mathbb{P}[br(T) > \lambda | T \text{ survives}] = 0$ as well. Next, take any $\lambda < \lambda^*$. Then,

$$\mathbb{P}[br(T) \leq \lambda] = \sum_{k=1}^{\infty} \mathbb{P}[br(T) \leq \lambda]^k p_k = \sum_{k=1}^{\infty} (1 - \mathbb{P}[br(T) > \lambda])^k p_k < 1$$

Importantly, notice that $\mathbb{P}[br(T) \leq \lambda]$ follows the same generating function as $\mathbb{P}[\text{extinction}]$ in Definition 4.16. Since $\lambda < \lambda^*$, $\mathbb{P}[br(T) > \lambda] > 0$. Thus, we have

$1 - \mathbb{P}[\text{br}(T) > \lambda] = \mathbb{P}[\text{br}(T) \leq \lambda] < 1$. Then, we can conclude that $\mathbb{P}[\text{br}(T) > \lambda] = \mathbb{P}[S]$.²²

So far, for $\mathbb{P}[\text{br}(T) \leq \lambda]$, we have not conditioned on anything. Now, let's condition on the branching process surviving.

$$\mathbb{P}[\text{br}(T) > \lambda | S] = \frac{\mathbb{P}[\text{br}(T) > \lambda, S]}{\mathbb{P}[S]} = \frac{\mathbb{P}[\text{br}(T) > \lambda]}{\mathbb{P}[S]} = 1$$

Second equality holds because the branching number being positive already implies S , and last equality comes from $\mathbb{P}[\text{br}(T) > x] = \mathbb{P}[S]$ from before. Then, we get

$$\mathbb{P}[\text{br}(T) > \lambda | S] = \begin{cases} 1, & \text{if } \lambda < \lambda^*. \\ 0, & \text{if } \lambda > \lambda^*. \end{cases} \quad (4.3)$$

Hence, $\text{br}(T) = \lambda^*$ a.s. conditioned on the survival T . Lastly, we will show that $\lambda^* \geq m$ a.s.

For any $mp > 1$, we know that

$$\begin{aligned} 0 &< \mathbb{P}[|T_o(\omega_p)| = \infty] \\ &= \mathbb{E}_{T \sim \mu_1} [\mathbb{P}[T_o(\omega_p) = \infty | T]] \\ &\leq \mathbb{E}_{T \sim \mu_1} [\mathbb{1}_{\{\mathbb{P}[T_o(\omega_p) = \infty | T] > 0\}}] \\ &= \mathbb{P}_{T \sim \mu_1} [\mathbb{P}[T_o(\omega_p) = \infty | T] > 0] \\ &= \mathbb{P}_{T \sim \mu_1} [p_c(T) < p] = \mathbb{P}_{T \sim \mu_1} [\text{br}(T) > 1/p] \end{aligned}$$

Second equality here uses the Law of Total Expectation whose proof uses Fubini's theorem. Fubini's Lemma's assumption (see Appendix) holds in this case since all values we are concerned with here are probabilities or indicators, hence everything is non-negative, and associated integrals are bounded by 1. Then, for any $p > \frac{1}{m}$ meaning that $\frac{1}{p} < m$, we have $\mathbb{P}[\text{br}(T) > 1/p] > 0$. Conditioning on the survival can only increase the branching number, then $\mathbb{P}[\text{br}(T) > 1/p | S] > 0$ too. Lastly, because conditioned on S , branching number is a single number λ^* almost surely, $\text{br}(T) = \lambda^* \geq m$ a.s. $\square$

One can generalize this proof to account for the multi-type branching process to get the following theorem. We will omit the proof here. The reader is encouraged to check Lyons's 1990 paper for the details [6].

Theorem 4.19. *Let m be the maximum eigenvalue of the mean matrix of a supercritical positive regular non-singular multi-type branching process. Given that the process does not become extinct, the branching number of the tree is m almost surely.*

Corollary 4.19.1. *A simple random walk on a Galton-Watson tree with mean offspring $m = \sum k p_k > 1$ is transient almost surely conditioned on the non-extinction of the tree.*

Remark 4.20. This result is interesting because the underlying graph is random. In the literature, this is called a RWRE (random walk in a random environment). Such a stochastic process combines the randomness of the environment with the randomness

²²This generating function has one or two roots and one of them is always 1 as described in Definition 4.16. $\mathbb{P}[\text{extinction}]$ corresponds to the smallest root. In our case, we assume that this branching process will survive with positive probability. Hence, there is one another root smaller than 1 corresponding to $\mathbb{P}[\text{extinction}]$.

of the walk which complicates the analysis.²³ Traditional properties like recurrence, transience, or the expected distance covered might change dramatically compared to standard random walks.

5. LOCALLY TREE-LIKE GRAPHS

Now, we will generalize the results we have proved for random trees to more complex graphs such as Erdős-Rényi and small-world networks that are locally tree-like using the theory of local convergence! Local convergence is a central technique in random graph theory that was first introduced by Benjamini and Schramm [1]. It describes the notion that a graph, seen from the perspective of a node chosen uniformly at random from all nodes, behaves like a certain limiting graph. For example, many random graphs converge into a branching process in the limit as we will see. Since random trees are generally simpler objects than most other random graphs, this means that to understand the random graph, it often suffices to understand the random tree instead! To introduce this theory, we will follow the notation in [4] closely.

5.1. Local Convergence of Graphs.

Definition 5.1. A *rooted graph* (G, o) is a graph G with a root node $o \in V(G)$. Let (G_1, o_1) and (G_2, o_2) be two rooted graphs. We say $(G_1, o_1) \simeq (G_2, o_2)$ or (G_1, o_1) and (G_2, o_2) are *isomorphic* to each other when there exists a bijection $f : V(G_1) \rightarrow V(G_2)$ such that:

- (1) $(u, v) \in E(G_1)$ if and only if $(f(u), f(v)) \in E(G_2)$ for all $u, v \in V(G_1)$.
- (2) $f(o_1) = o_2$.

We will denote $\mathcal{G}_*$ for the space of rooted graphs modulo isomorphism.

Definition 5.2. (Metric on rooted graphs $\mathcal{G}_*$) Let (G_1, o_1) and (G_2, o_2) be two rooted connected graphs, and $B_r^{(G)}(o)$ denote the (rooted) subgraph of (G, o) of all nodes at graph distance at most r away from o . The distance metric that allows $\mathcal{G}_*$ to be interpreted as a metric space is as follows:

$$d_{\mathcal{G}_*}((G_1, o_1), (G_2, o_2)) = \frac{1}{1 + R^*} \quad (5.1)$$

where

$$R^* = \sup\{r : B_r^{(G_1)}(o_1) \simeq B_r^{(G_2)}(o_2)\}$$

This metric space is nice, specifically, it is separable and Polish, allowing us to define a measure on it and have convergence in distribution, in probability, or almost surely on this space.

Definition 5.3. (Local convergence in probability) Let $(G_n)_{n \geq 1}$ with G_n denote a finite sequence of (possibly disconnected) random graphs with n nodes. Then, G_n converges locally in probability to (G, o) having law μ when

$$\mathbb{E}[h(G_n, o_n) | G_n] \xrightarrow{\mathbb{P}} \mathbb{E}_\mu[h(G, o)] \quad (5.2)$$

for every bounded and continuous function $h : \mathcal{G}_* \rightarrow \mathbb{R}$, where the expectation in the left-hand side is with respect to the randomness of the root o_n and the graph

²³Randomness in the environment could mean that the probabilities dictating movement from one point to another are random and either fixed once set (quenched randomness) or change dynamically as the walk progresses (annealed randomness). The decisions about where to move next are typically probabilistic and depend on the current position's properties in the environment.

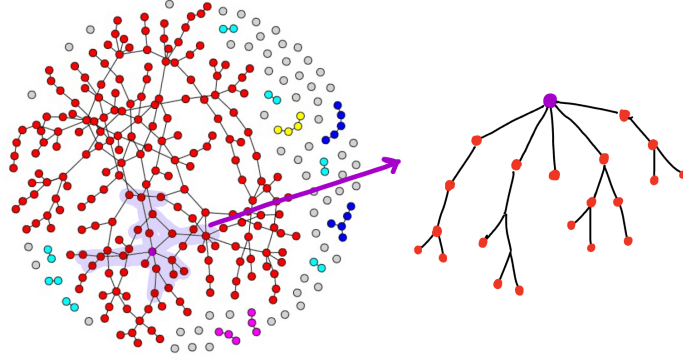


FIGURE 11. Erdős-Rényi is locally tree-like!

G_n and the expectation on the right-hand side is with respect to the randomness of the distribution. For this thesis' purposes, the limiting distribution is going to be deterministic (i.e. a single distribution $\bar{\mu}$ will have probability 1 in the limit). We will note this by $(G_n, o_n) \xrightarrow{\mathbb{P}} (G, o)$.

Remark 5.4. It is important to notice that here (G_n, o_n) 's and (G, o) are random variables on $\mathcal{G}_*$ with their respective laws (distributions or measures), and what's converging is their laws (distributions or measures). There is also the notion of local weak convergence of graphs corresponding to convergence in the distribution of their measures. However, since convergence in probability already implies convergence in the distribution, we omit that definition for the sake of brevity here.

We will use the theorem below to simplify the definition of local convergence in probability. The reader is referred to Chapter 2 of [4] for the proofs.

Theorem 5.5. (*Criterion for local convergence*) *Given a sequence of graphs $(G_n)_{n \geq 1}$, G_n converges locally in probability to $(G, o) \sim \mu$ if and only if*

$$p_r^{(G_n)}(H_*) = \frac{1}{|V(G_n)|} \sum_{v \in V(G_n)} \mathbb{1}_{\{B_r^{(G_n)}(v) \simeq H_*\}} \xrightarrow{\mathbb{P}} \mu(B_r^{(G)}(o) \simeq H_*) \quad (5.3)$$

for every rooted graph $H_* \in \mathcal{T}_*(r) \subseteq \mathcal{G}_*$ and any $\mathcal{T}_*(r)$ that satisfies $\mu(B_r^{(G)}(o) \in \mathcal{T}_*(r)) = 1$ for each integer $r \geq 0$.

Remark 5.6. This result is very powerful because, firstly, it reduces proving convergence to just **counting subgraph frequencies** and, secondly, given a guess for the limit graph distribution, it further reduces the proof of convergence to counting frequencies of subgraphs sampled from *only* this limit graph distribution.

One of the consequences of this theory is that it makes the notion that Erdős-Rényi (defined when we introduced percolation) is *locally tree-like* rigorous.

Theorem 5.7. (*Local convergence of Erdős-Rényi random graph*) *Fix $\lambda > 0$. $G(n, \lambda/n)$ converges locally in probability to a Poisson branching process with mean offspring λ .*

Remark 5.8. We will not explain this proof in detail since similar versions of the arguments here are used and explained in depth to prove the local convergence of small-world

networks in the last subsection 5.3. Since the proof for small-world networks does not exist in the literature to our knowledge, we wanted to give further information there. The reason behind the similarity of the proofs is that, in the limit, the random short-cuts of a small-world network (defined in 5.3) behave like the edges of an Erdős-Rényi graph. Curious readers can refer to Chapter 2 in [4] if they want to see a more rigorous treatment of the proof below.

Proof. Let $G_n = G(n, \lambda/n)$ and $Poi(\lambda)$ be the Poisson branching process with mean λ . Using the criterion given by Theorem 5.5, we prove the convergence of subgraph proportions in 5.3 only for the r -neighborhood around the root of rooted trees that can be sampled with a positive probability from $Poi(\lambda)$. Let's fix one such r -neighborhood of a tree: t . We need to show that, for every such tree t and all integers $r \geq 0$,

$$p^{(G_n)}(t) = \frac{1}{n} \sum_{u \in [n]} \mathbf{1}_{\{B_r^{(G_n)}(u) \simeq t\}} \xrightarrow{\mathbb{P}} \mu(B_r^{(G)}(o) \simeq t)$$

where μ of (G, o) is the distribution of a $Poi(\lambda)$ branching process.

One can order this t as well as G_n and G starting from their root through a Breadth-First Search process, breaking ties arbitrarily. We make this explicit by writing $\bar{B}_r^{(G)}(o)$ for the ordered version of $B_r^{(G)}(o)$ and $\bar{t}$ for the ordered version of t . Therefore, we can write $\bar{B}_r^{(G)}(o) = \bar{t}$ to indicate that the two ordered trees $\bar{B}_r^{(G)}(o)$ and $\bar{t}$ agree. Then,

$$\frac{1}{n} \sum_{u \in [n]} \mathbf{1}_{\{B_r^{(G_n)}(u) \simeq t\}} = \frac{\#(t)}{n} \sum_{u \in [n]} \mathbf{1}_{\{B_r^{(G_n)}(u) = t\}} \xrightarrow{\mathbb{P}} \#(t) \mu(B_r^{(G)}(o) = t) = \mu(B_r^{(G)}(o) \simeq t)$$

where $\#(t)$ gives the number of different ways one can relabel t breaking ties differently.

Since $G \sim Poi(\lambda)$, one can compute

$$\mu(\bar{B}_r^{(G)}(o) = t) = \prod_{i \in [t]: \text{dist}(\bar{o}, a_i) < r} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

where x_i denotes the number of offspring of v_i in t . What's left to show is that

$$\frac{1}{n} \sum_{u \in [n]} \mathbf{1}_{\{B_r^{(G_n)}(u) = t\}} \xrightarrow{\mathbb{P}} \prod_{i \in [t]: \text{dist}(\bar{o}, a_i) < r} \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

We will start by showing the convergence of the first moment, namely

$$\mathbb{E} \left[\frac{1}{n} \sum_{u \in [n]} \mathbf{1}_{\{B_r^{(G_n)}(u) = t\}} \right] = \mathbb{P}(\bar{B}_r^{(G_n)}(1) = t)$$

where the equality is due to symmetry.

Let X_i be the number of offspring of node i in G_n can be so that $B_r^{(G_n)}(u)$ is still a tree isomorphic to t up until node i based on the information we know after we are done exploring $i - 1$ -th node in the Breadth First Search process to order G_n . Notice

$$X_i \sim \text{Bin}(n_{i-1}, \lambda/n),$$

where n_{i-1} equals to $n - s_{i-1} - (i - 1)$, s_{i-1} is the number of active vertices we have found but not yet explored, and $i - 1$ is the number of already explored edges once we

explored $i - 1$ -th node. We are subtracting s_{i-1} and $i - 1$ from n because we do not want node i to be part of a loop. Then,

$$\mathbb{P}(\overline{B}_r^{(G_n)}(1) = t) = \prod_{i \in [t]: \text{dist}(\emptyset, a_i) < r} \mathbb{P}(\text{Bin}(n_i, \lambda/n) = x_i) \times \prod_{i \in [t]} \left(1 - \frac{\lambda}{n}\right)^{s_{i-1}}$$

where $\prod_{i \in [t]} \left(1 - \frac{\lambda}{n}\right)^{s_{i-1}}$ ensures that node i does not create any loop with already found nodes. Therefore, since $\mathbb{P}(\text{Bin}(n_i, \lambda/n) = x_i) \rightarrow e^{-\lambda \frac{\lambda^{x_i}}{x_i!}}$ and s_i , λ , and $-s_{i-1} - (i - 1)$ are all constants with respect to n for any locally finite G_n (which is the case here), we have

$$\mathbb{P}(\overline{B}_r^{(G_n)}(1) = t) \rightarrow \prod_{i \in [t]: \text{dist}(\emptyset, a_i) < r} e^{-\lambda \frac{\lambda^{x_i}}{x_i!}} = \mu(\overline{B}_r^{(G)}(o) = t) = \mu(\overline{B}_r^{(G)}(o) = t).$$

The proof of the convergence of the second moment is the same as the one for small-world networks in Section 5.3 except for the calculation of $\mathbb{E}[|B_r^{(G_n)}(v)|]$ where v is a node chosen uniformly at random from G_n . In the case of Erdős-Rényi with n nodes, the expected degree of a node is λ . Thus, by induction on k , one can show that $\mathbb{E}[|B_r^{(G_n)}(v)|]$ is bounded above by $1 + \lambda + \dots + \lambda^r = \frac{\lambda^{r+1} - 1}{\lambda - 1}$ which is a constant with respect to n and the result follows (refer to Second Moment subsection in 5.3 for the details). $\square$

5.2. Phase Transition in Erdős-Rényi. As a graph grows, understanding its global properties can become intractable, but this theory will allow us to estimate such global-level properties by examining local neighborhoods. In particular, we will explain why the size of the giant component in Erdős-Rényi, while seemingly global, ends up being a local function of the graph and recover the phase transition threshold for Erdős-Rényi as a result. Let's define this property formally.

Definition 5.9. A *giant component* in a graph is a connected subgraph containing a positive fraction of the total nodes as the graph size grows. In other words, the size of the giant is $\theta(n)$ where n is the number of nodes. For a graph G with n nodes, the *phase transition* refers to the critical point p_c ²⁴ for p where a giant component emerges when one thins (or percolates) the edges of G with probability $1 - p$. If $p < p_c$, all components are small, i.e. their size is $o(n)$, but, if $p > p_c$, a giant component appears with high probability.

$\mathcal{C}_{(i)}^G$ will denote i^{th} largest component (which can be a giant component) with ties broken arbitrarily in the graph G . The largest component $\mathcal{C}_{(1)}^G$ will be referred as $\mathcal{C}_{\max}^G$ at times. Lastly, $\mathcal{C}_v^G$ will be the connected component containing the node v . If the graph we are talking about is obvious, we will drop G . Now, we are ready to characterize the giant in terms of its limit.

Theorem 5.10. (*Upper bound on the giant*) Assume that G_n converges locally in probability to $(G, o) \sim \mu$. As before, $\zeta = \mu(|\mathcal{C}_{\max}^G| = \infty)$ for the survival probability of the limiting graph (G, o) . Then, for every $\epsilon > 0$ fixed,

$$\mathbb{P}(|\mathcal{C}_{\max}^{G_n}| \leq n(\zeta + \epsilon)) \rightarrow 1.$$

In particular, when $\zeta = 0$, then,

$$\frac{|\mathcal{C}_{\max}^{G_n}|}{n} \xrightarrow{\mathbb{P}} 0$$

²⁴This is different than percolation threshold.

meaning that there can only be a giant when the local limit has a positive survival probability.

Proof. Since we want to prove a result holds in probability, we will first introduce a probabilistic little ‘oh’ notation that will be useful: if $\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}(|X_{n,k}| > \epsilon) = 0$, then we say $X_{n,k} = o_{k,\mathbb{P}}(1)$.

Now, assume G_n converges in probability to (G, o) . Second, let’s define $V_{\geq k} = \sum_{v \in V(G_n)} \mathbb{1}_{\{|C_v^{G_n}| \geq k\}}$ for G_n and $\zeta_{\geq k} = \mathbb{P}(|C_o^G| \geq k)$ for G . Because the fraction of nodes in G_n that are part of each subgraph with more than k nodes converge in probability to the corresponding fraction for (G, o) ,

$$\frac{V_{\geq k}}{n} = \frac{1}{n} \sum_v \mathbb{1}_{\{|C_v^{G_n}| \geq k\}} = \sum_{i \geq 1} |C_{(i)}^{G_n}| \mathbb{1}_{\{|C_{(i)}| \geq k\}} \xrightarrow{\mathbb{P}} \zeta_{\geq k}$$

Note that $\zeta_{\geq k} \rightarrow \zeta$, so we have

$$\frac{V_{\geq k}}{n} = \frac{1}{n} \sum_v \mathbb{1}_{\{|C_v^{G_n}| \geq k\}} = \sum_{i \geq 1} |C_{(i)}^{G_n}| \mathbb{1}_{\{|C_{(i)}| \geq k\}} = \zeta + o_{k,\mathbb{P}}(1) \quad (5.4)$$

Furthermore, we can choose large enough K such that $\zeta \geq \zeta_{\geq K} - \epsilon/2$ for any ϵ . Now, observe

$$\mathbb{P}(C_{\max} \geq n(\zeta + \epsilon)) \leq \mathbb{P}(V_{\geq k} \geq n(\zeta + \epsilon)) \leq \mathbb{P}(V_{\geq k} \geq n(\zeta_{\geq K} - \epsilon/2)) = o(1)$$

as K and n get larger by equation 5.4. Since ϵ is arbitrary, $n\zeta$ is an upper bound. $\square$

This theorem has a significant consequences for Erdős-Rényi $G(n, \frac{\lambda}{n})$. Specifically, it shows that the size of the giant is bounded by $\zeta \cdot n$ with high probability where ζ is the survival probability of $Poi(\lambda)$ branching process. Even better, we will be able to give an exact order of the giant in Erdős-Rényi via the next theorem!

Theorem 5.11. *Assume that G_n converges locally in probability to $(G, o) \sim \mu$. Say $\zeta = \mu(|C_{\max}^G| = \infty)$ and*

$$F_{k,n} = \frac{1}{n^2} \mathbb{E}[\#\{(x, y) \in V(G_n) \times V(G_n) : |C(x)|, |C(y)| \geq k, x \leftrightarrow y\}]$$

be the expected fraction of disconnected pairs belonging to connected components with more than k nodes in G_n . Assume $\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} F_{k,n} = 0$. Then, with $C_{\max} = C_{(1)}^{G_n}$ and $C_{(i)} = C_{(i)}^{G_n}$, we have

$$\frac{|C_{\max}|}{n} \xrightarrow{\mathbb{P}} \zeta = \mu(|C(o)| = \infty), \frac{|C_{(2)}|}{n} \xrightarrow{\mathbb{P}} 0$$

Proof. The previous theorem gives us the upper bound, and now we will prove the lower bound. To do so, we first prove an important lemma.

Lemma 5.12. *If $\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} F_{k,n} = 0$, then*

$$\frac{1}{n^2} \sum_{i \geq 1} |C_{(i)}|^2 \mathbb{1}_{\{|C_{(i)}| \geq k\}} = \zeta^2 + o_{k,\mathbb{P}}(1)$$

Proof. We will use the second-moment method for this proof. One can rewrite $\zeta + o_{k,\mathbb{P}}(1) = \frac{1}{n} \sum_{i \geq 1} |C_{(i)}| \mathbb{1}_{\{|C_{(i)}| \geq k\}}$. Now, let’s square both sides:

$$\zeta^2 + o_{k,\mathbb{P}}(1) = \frac{1}{n^2} \sum_{i \geq 1} |C_{(i)}|^2 \mathbb{1}_{\{|C_{(i)}| \geq k\}} + \frac{1}{n^2} \sum_{i \geq 1} \sum_{j \neq i} |C_{(i)}| |C_{(j)}| \mathbb{1}_{\{|C_{(i)}| \geq k\}} \mathbb{1}_{\{|C_{(j)}| \geq k\}}$$

The second term equals to $F_{k,n}$, so it is 0 in the limit. $\square$

Going back to the original theorem, we get a lower bound:

$$\zeta^2 + o_{k,\mathbb{P}}(1) = \frac{1}{n^2} \sum_{i \geq 1} |\mathcal{C}_{(i)}|^2 \mathbb{1}_{\{|\mathcal{C}_{(i)}| \geq k\}} \leq \frac{|\mathcal{C}_{(max)}|}{n^2} \sum_{i \geq 1} |\mathcal{C}_{(i)}| \mathbb{1}_{\{|\mathcal{C}_{(i)}| \geq k\}} = \frac{|\mathcal{C}_{(max)}|}{n} (\zeta + o_{k,\mathbb{P}}(1))$$

Lastly, dividing each side by ζ , we get $\zeta + o_{k,\mathbb{P}}(1) \leq \frac{|\mathcal{C}_{(max)}|}{n}$. Since k is arbitrary here, we can conclude that

$$\frac{|\mathcal{C}_{max}|}{n} \xrightarrow{\mathbb{P}} \zeta \quad (5.5)$$

For the second largest component, by equations 5.4 and 5.5, we get $\frac{|\mathcal{C}_{(2)}|}{n} \xrightarrow{\mathbb{P}} 0$. $\square$

Remark 5.13. Intuitively, $\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} F_{k,n} = 0$ ensures that the distribution of the graph in the local limit controls the size of the giant because it guarantees that there cannot be multiple disjoint arbitrarily large components. Because the convergence of the size of the giant requires this extra condition, the giant is called “almost” local in the literature.

Importantly, Erdős-Rényi satisfies the assumptions of Theorem 5.11. For the proof, one should first notice the simplification below.

Theorem 5.14. *Assume that G_n converges locally in probability to $(G, o) \sim \mu$. Say $\zeta = \mu(|\mathcal{C}_{max}^G| = \infty)$. Then, we know that condition $\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} F_{k,n} = 0$ if*

$$\lim_{r \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n^2} |\#\{(x, y) : |\partial B_r^{G_n}(x)|, |\partial B_r^{G_n}(y)| \geq r, x \leftrightarrow y\}| = 0, \quad (5.6)$$

provided that there exists $r = r_k \rightarrow \infty$ such that, as $k \rightarrow \infty$,²⁵

$$\mu(|\mathcal{C}_o^G| \geq k, |\partial B_r^G(o)| < r) \rightarrow 0, \quad \mu(|\mathcal{C}_o^G| < k, |\partial B_r^G(o)| \geq r) \rightarrow 0 \quad (5.7)$$

We have already mentioned that Erdős-Rényi converges in probability to the Poisson(λ) branching process. Before giving the proof of this theorem, we will motivate it by showing that a Poisson(λ) branching process satisfies assumptions 5.7. Let $G \sim \mu$ be a Poisson(λ) branching process. For each k , take $r_k = k$. Clearly, $\mu(|\mathcal{C}_o^G| < k, |\partial B_r^G(o)| \geq r) = 0$ for any k . To prove $\mu(|\mathcal{C}_o^G| \geq k, |\partial B_r^G(o)| < r) \rightarrow 0$, we will take advantage of Kesten Stigum Theorem.

Theorem 5.15. (*Kesten-Stigum Theorem*) *For a branching process with i.i.d. offspring X having mean $\mu = \mathbb{E}[X] > 1$, $\mu^{-n}Z_n$ converges almost surely to some finite random variable W with probability 1. Furthermore, conditioned on non-extinction, W is positive with probability 1.*²⁶

If the branching process goes extinct, $\mu(\mathcal{C}_o^G \geq k) \rightarrow 0$ as $k \rightarrow \infty$. If it doesn't, conditioned on non-extinction, $|\partial B_r^G(o)| = Z_r \sim W\mu^r \rightarrow \infty$ as r grows since $W > 0$ by Kesten-Stigum. Thus, $\mu(|\partial B_r^G(o)| < r) \rightarrow 0$. Hence, for Erdős-Rényi, we can actually use condition (4.23) instead. Now let's prove Theorem 5.14.

²⁵Qualitatively, assumptions 5.7 corresponds to the tree not being too tall or fat.

²⁶The proof relies on $M_n = \mu^{-n}Z_n$ being a Martingale and then the Martingale Convergence Theorem (see Appendix) for almost sure convergence. By properties of martingales, we know that $1 = \mathbb{E}[M_0] = \mathbb{E}[M_n] = \mathbb{E}[W]$. Therefore, W is finite a.s.

Proof. Denote

$$P_k = \{\#(x, y) : |\mathcal{C}_x^{G_n}|, |\mathcal{C}_y^{G_n}| \geq k, x \leftrightarrow y\}$$

$$P'_r = \{\#(x, y) : |\partial B_r^{G_n}(x)|, |\partial B_r^{G_n}(y)| \geq r, x \leftrightarrow y\}$$

As before, define

$$V_{<r, \geq k} = \sum_{v \in V(G_n)} \mathbb{1}_{\{|\partial B_r^{G_n}(v)| < r, |\theta(v)| \geq k\}}$$

$$V_{\geq r, < k} = \sum_{v \in V(G_n)} \mathbb{1}_{\{|\partial B_r^{G_n}(v)| \geq r, |\mathcal{C}_v^{G_n}| \geq k\}}$$

Then,

$$|P_k - P'_r| \leq 2n[V_{<r, \geq k} + V_{\geq r, < k}]$$

Therefore, by local convergence in probability, as $n \rightarrow \infty$,

$$\frac{1}{n^2}|P_k - P'_r| \leq \frac{2}{n}[V_{<r, \geq k} + V_{\geq r, < k}]$$

$$\xrightarrow{\mathbb{P}} 2 \cdot \mu(|\mathcal{C}_o^G| \geq k, |\partial B_r^G(o)| < r) + 2 \cdot \mu(|\mathcal{C}_o^G| < k, |\partial B_r^G(o)| \geq r)$$

Take $r = r_k$. Then, the ending result vanishes as $k \rightarrow \infty$ by assumption 5.7, so $\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{n^2}|P_k - P'_r| = 0$.

Notice for any k , $\lim_{n \rightarrow \infty} \frac{1}{n^2}|P_k - P'_r| \leq \frac{2}{n}[V_{<r, \geq k} + V_{\geq r, < k}] \leq \frac{2}{n}(n + n) = 4$. Then, by the Dominated Convergence Theorem (see Appendix),

$$\lim_{k \rightarrow \infty} \mathbb{E}[\lim_{n \rightarrow \infty} \frac{1}{n^2}|P_k - P'_r|] = \mathbb{E}[\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{n^2}|P_k - P'_r|] = 0.$$

Similarly, for any n , $|\frac{1}{n^2}|P_k - P'_r|| \leq \frac{1}{n^2}2n^2 = 2$. By the Dominated Convergence Theorem,

$$\lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \mathbb{E}[\frac{1}{n^2}|P_k - P'_r|] = \lim_{k \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{n^2}\mathbb{E}[|P_k - P'_r|] = 0.$$

All in all,

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n^2}\mathbb{E}[P_k] \leq \lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{1}{n^2}\mathbb{E}[|P_k - P'_r|] + \frac{1}{n^2}\mathbb{E}[P'_r] = 0, \quad (2.6.47)$$

where last term vanishes because $r := r_k \rightarrow \infty$ when $k \rightarrow \infty$. $\square$

We will not get into the full proof of why Erdős-Rényi satisfies this new condition 5.6 in Theorem 5.14. The reader is referred to [9] for the details. On the high level, they first notice that if one chooses $o_1, o_2 \in V(G_n)$ uniformly at random, $\mathbb{P}(|\partial B_r^{G_n}(o_1)|, |\partial B_r^{G_n}(o_2)| \geq r, o_1 \leftrightarrow o_2)$ gives the same expression as the expression 5.6. Secondly,

$$\mathbb{P}(|\partial B_r(o_1)|, |\partial B_r(o_2)| \geq r, o_1 \leftrightarrow o_2)$$

$$= \sum_{b_0^{(1)}, b_0^{(2)}, s_0^{(1)}, s_0^{(2)}} \mathbb{P}(o_1 \leftrightarrow o_2 | |\partial B_r(o_i)| = b_0^{(i)}, B_{r-1}(o_i) = s_0^{(i)}, i \in \{1, 2\})$$

$$\mathbb{P}(|\partial B_r(o_i)| = b_0^{(i)}, B_{r-1}(o_i) = s_0^{(i)}, i \in \{1, 2\}).$$

Then the aim is to show that, for every $\varepsilon > 0$, we can find $r = r_\varepsilon$ such that, for every $b_0^{(1)}, b_0^{(2)} \geq r$ and $s_0^{(1)}, s_0^{(2)}$ fixed,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(o_1 \leftrightarrow o_2 | |\partial B_r(o_i)| = b_0^{(i)}, B_{r-1}(o_i) = s_0^{(i)}, i \in \{1, 2\}) \leq \varepsilon.$$

Conditioned on $|\partial B_r(o_i)| = b_0^{(i)}$, $B_{r-1}(o_i) = s_0^{(i)}$, $i \in \{1, 2\}$, they notice that

$$|\partial B_{r+1}(o_1) \setminus B_r(o_2)| \sim \text{Bin}(n, p),$$

where

$$n = n - s_0^{(1)} - s_0^{(2)},$$

$$p = 1 - \left(1 - \frac{\lambda}{n}\right)^{b_0^{(1)}}.$$

(One can get a similar expression for $B_{r+k}(o_2)$) Here, $1 - \left(1 - \frac{\lambda}{n}\right)^{b_0^{(1)}}$ makes sense since it gives the probability at least one of the nodes in $B_r(o_1)$ will be connected to each node outside of $B_r(o_1)$ and $B_r(o_2)$, which suffices to increase $|\partial B_{r+1}(o_1) \setminus B_r(o_2)|$. Then, they use the following lemma to show that the boundary $|\partial B_{r+1}(o_1)|$ is close to $\lambda|\partial B_r(o_1)|$ for r and n large with high probability so that the boundary grows by a factor $\lambda > 1$, eventually giving $|\partial B_{r+k}(o_1)| \approx \lambda^k |\partial B_r(o_1)|$. Thus, in roughly $a \log_\lambda n$ steps, the boundary will have expanded to n^a vertices.

Lemma 5.16. *Let $X \sim \text{Bin}(n, p)$. Then, for every $\delta > 0$,*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \delta \mathbb{E}[X]) \leq 2 \exp\left(-\frac{\delta^2 \mathbb{E}[X]}{2(1 + \delta/3)}\right).$$

Then, they grow $|\partial B_{r+k}(o_1)|$ for $k \geq 1$ until $|\partial B_{r+k}(o_1)|$ is very large. Then, outside of $B_{r+k}(o_1)$ for the appropriate k , they grow $|\partial B_{r+k}(o_2)|$ for $k \geq r$ until also $|\partial B_{r+k}(o_2)|$ is large too. In the end, the boundaries are so large that it becomes very likely the resulting boundaries intersect! Recall the proof of Theorem 4.7 had a very similar story with the twist that the clusters there were infinite. As a corollary of this result, we recover the phase transition for the Erdős-Rényi.

Corollary 5.16.1. *(Phase Transition of the Erdős-Rényi) The threshold function for the emergence of a giant component of size $\theta(n)$ for $G(n, p)$ is $1/n$. If $p > 1/n$, the ratio of the largest component's size to the size of the whole graph will tend to ζn as $n \rightarrow \infty$ where ζ is the survival probability of the underlying Poisson(λ) branching process.*

This particular proof of the phase transition threshold for Erdős-Rényi is interesting because it also explains why the phase transition threshold is connected to the critical percolation threshold of the Poisson branching process.

5.3. Small World Networks. Now, we will apply the local convergence toolkit to another local-tree-like graph: small-world networks! Specifically, we will show that small-world networks converge in probability to *multi-type branching process* introduced in Definition 4.17. To be able to account for different types of nodes, we will assign each node a mark, corresponding to its type, and adapt our theory to account for marked graphs as follows.

Definition 5.17. We will consider *marked* rooted graphs to be a rooted graph (G, o) together with a set $M(G)$ such that M maps from $V(G)$ and $E(G)$ to a complete separable metric space.

The distance metric that allows $\mathcal{G}_*$ to be interpreted as a metric space is as follows:

$$d_{\mathcal{G}_*}((G_1, o_1, M_1), (G_2, o_2, M_2)) = \frac{1}{1 + R^*} \quad (5.8)$$

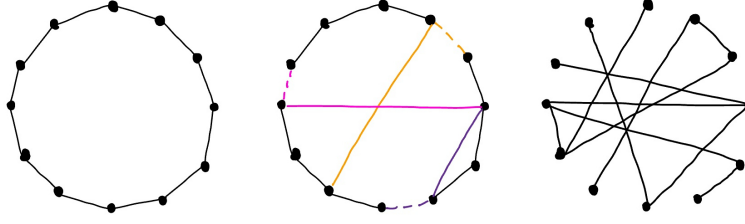


FIGURE 12. (left) $\phi = 0$: 1-dimensional ring lattice (middle) $0 < \phi < 1$: small-world network where colored edges are the “shortcuts” (right) $\phi = 1$ random graph Note: Dashed edges correspond to the original location of shortcuts before they were rewired

where

$$\begin{aligned} R^* &= \sup\{r : B_r^{(G_1)}(o_1) \simeq B_r^{(G_2)}(o_2), \\ &\exists \pi \text{ such that } M_1(u) = M_2(\pi(u)) \text{ for all } u \in V(B_r^{(G_1)}(o_1)), \\ &M_1((u, v)) = M_2(\pi((u, v))) \text{ for all } (u, v) \in E(B_r^{(G_1)}(o_1))\} \end{aligned}$$

The latter two conditions for mark equalities will be assumed when $B_r^{(G_1)}(o_1) \simeq B_r^{(G_2)}(o_2)$ is true from now on for simplicity. As before, one can use Definition 5.3 (Local convergence in probability) and Theorem 5.5 (Criterion for local convergence) with this new metric $d_{\mathcal{G}_*}$ in 5.8 and with $\simeq$ accounting for the corresponds of marks in each graph as well. Let’s get started!

A small-world network is a graph with a small average distance between nodes despite the high clustering coefficient. Due to these properties, small-world networks inspired many network theory applications in real life. Watts and Strogatz proposed a procedure to build such graphs as follows [10]:

- (1) Start with a n -cycle and a root o .
- (2) Direct all edges in a clockwise direction and rewire each edge’s outgoing direction to a uniformly selected node in the graph with probability ϕ . Finally, make the graph undirected.

We will call the rewired edges “shortcuts” and non-rewired ones “ring edges.” We will refer to the random small world network sampled through this procedure with parameters n and ϕ as $SW(n, \phi)$ (See Figure 12). We will prove that the distribution of a local neighborhood of a node selected uniformly at random from $SW(n, \phi)$ as $n \rightarrow \infty$ converges in probability to one selected from a multi-type branching process. Intuitively, a tree as the local limit makes sense since the probability that a shortcut is going to create a cycle of a small length goes to zero as the graph gets larger, so there will be no cycles in the local neighborhood of a node with high probability. See Figure 14 to view a local neighborhood of a small-world network as a tree. Secondly, this tree should have offspring of multiple types because, for example, the distribution of shortcuts would differ from the distribution of ring edges. Furthermore, a node v can have two types of ring edges as offspring: left (anti-clockwise) edges and right (anti-clockwise) edges. Similarly, it can have two different types of shortcuts: a shortcut that rewired *to* it (i.e. an edge that connected to it after rewiring) and a shortcut that rewired *from* it (i.e. an edge that was connected to node v before being rewired and is

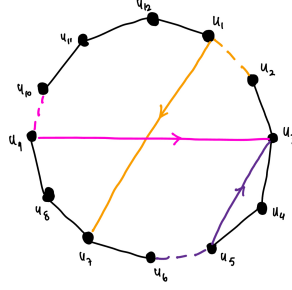


FIGURE 13. The arrows on the shortcuts give the direction in which they were rewired: u_3 has two shortcuts rewired *to* it; u_1 has one shortcut rewired *from* it. u_3 has two offspring on the ring: u_2 is the left one whereas u_4 is the right one.

still connected to it). See Figure 13 to understand *to* and *from* directions of shortcuts and the orange marks in Figure 14 for an example.

All in all, there are 4 different types:

- (1) A shortcut rewired *to* v
- (2) Right ring node
- (3) Left ring node
- (4) A shortcut rewired *from* v

We will refer to these types as *marks*.

Theorem 5.18. *Marked rooted small world network $SW(n, \phi)$ with $0 < \phi < 1$ converges locally in probability to a marked multi-type branching process with 4 node marks whose offspring distribution are as follows*

- (1) $\mu_1(x_1, x_2, x_3, x_4) = \mathbb{P}[X_1 = x_1] \mathbb{1}_{\{x_1+x_2=1\}} \mathbb{P}[X_3 = x_3] \mathbb{P}[X_4 = x_4]$
- (2) $\mu_2(x_1, x_2, x_3, x_4) = \mathbb{P}[X_1 = x_1] \mathbb{1}_{\{x_1+x_2=1\}} \mathbb{1}_{\{x_3=0\}} \mathbb{P}[X_4 = x_4]$
- (3) $\mu_3(x_1, x_2, x_3, x_4) = \mathbb{1}_{\{x_1=0, x_2=0\}} \mathbb{P}[X_3 = x_3] \mathbb{P}[X_4 = x_4]$
- (4) $\mu_4(x_1, x_2, x_3, x_4) = \mathbb{1}_{\{x_1=0, x_2=0\}} \mathbb{P}[X_3 = x_3] \mathbb{P}[X_4 = x_4]$

where $\mu_j(x_1, x_2, x_3, x_4)$ corresponds to the probability that a mark (j) node has x_1, x_2, x_3, x_4 many marks (1), (2), (3), and (4) nodes as offspring respectively, and

- $X_1 \sim \text{Ber}(\phi)$,
- $X_3 \sim \text{Ber}(1 - \phi)$,
- and $X_4 \sim \text{Poi}(\phi)$

are independent random variables. The root o is always marked (1). Let's call this model $MBP(\phi)$.

Proof. Let $G_n = SW(n, \phi)$ and $(G, o) = MBP(\phi)$. Using the criterion given by Theorem 5.5, we prove the convergence of subgraph proportions in 5.3 only for the r -neighborhood around the root of rooted *marked trees* that can occur with a positive probability according to the multi-type branching process above for any r . Fix one such marked tree $\mathbf{t} \sim MBP(\phi)$. Let's order vertices of (G, o) and $\mathbf{t}$ following a Breadth-First Search process from the root,²⁷ prioritizing nodes with smaller marks in the case of ties. Let these ordered and marked trees be $\bar{G}$ and $\bar{\mathbf{t}}$. Notice that this ordering is not unique.

²⁷Recall that each node has finitely many offspring almost surely in the process outlined above. Hence, this breadth-first search process is guaranteed to terminate.

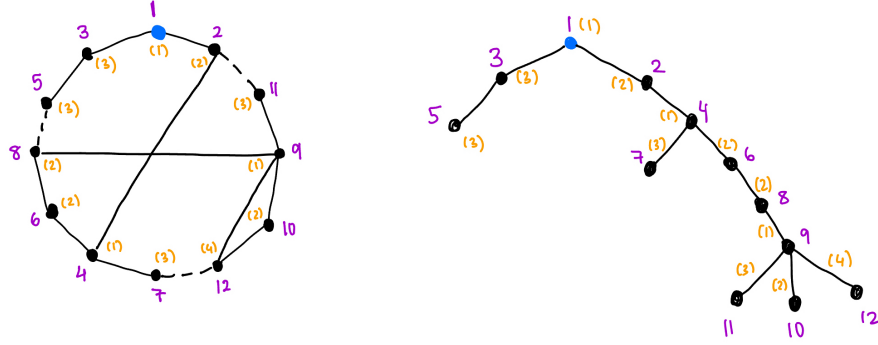


FIGURE 14. (left) Ordered in purple and marked in orange $G_{12} = SW(12, \phi) = B_6^{\overline{G}_{12}}(o)$ (right) viewing same graph as a tree (blue node corresponds to the root)

However, any other ordering of the graph $\bar{t}$ would have given an isometric graph to $\mathbf{t}$ since the underlying structure is not changed. Thus, we get the following identity²⁸:

$$\mu(B_r^{(G)}(o) \simeq t) = \#(t) \cdot \mu(B_r^{\overline{G}}(o) = \bar{t})$$

where $\#(t)$ is the number of different ways we could order this marked tree. Then,

$$\mu(B_r^{\overline{G}}(o) = \bar{t}) = \prod_{\substack{i \in [t] \\ \text{dist}_G(o, v_i) < r}} \sum_{j=1}^4 \mathbb{1}_{\{\mu(v_i)=j\}} \mu_j(x_{i,1}, x_{i,2}, x_{i,3}, x_{i,4}) \quad (5.9)$$

where $x_{i,j}$ is the number of j marked offspring of node v_i . We want to show that

$$p_r^{(G_n)}(t) \xrightarrow{\mathbb{P}} \mu(B_r^{(G)}(o) \simeq t) \quad (5.10)$$

To evaluate $p_r^{(G_n)}(t)$, we first need to order and mark $B_r^{(G_n)}(o)$ similar to G and h .

Definition 5.19. Let $B_r^{\overline{G}_n}(o)$ be the marked and ordered r -neighborhood of (G_n, o_n) (where $G_n = SW(n, \phi)$ and $o := o_n$ is randomly selected root) obtained by ordering vertices of G_n following a BFS (Breadth-First Search) process for r steps from the root o and marking a node v with:

- (1) if BFS reaches v first through a shortcut rewired *to* v
- (2) if BFS reaches v first through a ring edge in the clockwise direction of its parent
- (3) if BFS reaches v first through a ring edge in the anti-clockwise direction of its parent
- (4) if BFS reaches v first through through a shortcut rewired *from* v

as the BFS continues. In the case of ties, BFS should prioritize nodes with smaller marks as before. Note that the root o is always marked with (1).²⁹ See Figure 14 for an example with $n = 12$.

Although the markings of the nodes are unique, the ordering of the nodes is not. In the event that $B_r^{\overline{G}_n}(o) = \bar{t}$, we know that both graphs are ordered the same way,

²⁸‘=’ indicates ordered equivalence.

²⁹Notice that each node of $SW(n, \phi)$ will have at most two ring edges and on expectation $1\phi + n \cdot \phi \cdot \frac{1}{n} = 1$ shortcuts connected to it. Therefore, the degree of any node in $B_r^{(G_n)}(o)$ is finite almost surely, and so the BFS process will terminate almost surely.

which is only possible if $B_r^{(G_n)}(v)$ is the same marked tree. Thus, $\#(t) = \#(B_r^{(G_n)}(v))$ given $B_r^{(\overline{G_n})}(v) = \bar{t}$ and we get

$$p_r^{(G_n)}(t) = \frac{\#(t)}{n} \sum_{i \in [n]} \mathbb{1}_{\{B_r^{(\overline{G_n})}(v_i) = \bar{t}\}}$$

As before, the conditions for mark equalities will be assumed to hold when $B_r^{(G_1)}(o_1) = B_r^{(G_2)}(o_2)$ is true.

Given these identities, we will use the Second Moment Method to prove the convergence. Let $p_r^{\overline{G_n}}(t) = \frac{p_r^{(G_n)}(t)}{\#(t)}$.

$$\mathbb{E}[p_r^{\overline{G_n}}(t)] = \frac{1}{n} \sum_{i \in [n]} \mathbb{P}(B_r^{(\overline{G_n})}(v_i) = \bar{t}) = \mathbb{P}(B_r^{(\overline{G_n})}(v_1) = \bar{t}) \quad (5.11)$$

where the first equality follows from the linearity of expectation and the second from the symmetry of the distribution of neighborhoods. Before writing the closed form for $\mathbb{P}(B_r^{(\overline{G_n})}(v_1) = \bar{t})$, first notice that beyond getting the degree of the nodes right, we need to ensure that $B_r^{(\overline{G_n})}(v_1)$ has no cycles to ensure equality to tree $\bar{t}$.

Now, let's introduce some notation to keep things clean. Let

$$s_i = n - \sum_{k=1}^{i-1} x_{k,1} + x_{k,2} + x_{k,3} + x_{k,4}$$

be the number of nodes that have not yet been explored in the i -th step of the breadth-first search process, and

$$l_i = \sum_{k=1}^{i-1} \mathbb{1}_{\{M_n(v_k)=3, x_{k,3}=0\}}$$

be the number of anti-clockwise edges of already explored nodes that were rewired away from them.³⁰ Since G_n is locally finite (no node has infinitely many offspring), $\lim_{n \rightarrow \infty} \frac{s_i}{n} = 1$ and l_i is bounded for each i .

Further, let n_i be the number of nodes in G_n that mark (1) off-spring of v_i could be so that there will be no cycles in $B_r^{(\overline{G_n})}(v_1)$ created by the shortcut rewired from v_i to this mark (1) offspring and $m_{i,k}$ be the number of nodes in G_n that k -th mark (4) offspring of v_i ³¹ could be so that there will be no cycles in $B_r^{(\overline{G_n})}(v_1)$ created by the shortcut rewired from this mark (4) offspring to v_i , both disregarding shortcuts that could be encountered in the Breadth-First Search process after step i (we will account for them as we go in the breadth-first search process).

For this proof, it will suffice to give lower and upper bounds n_i and $m_{i,k}$. Obviously, $n_i, m_{i,k} \leq n$. For the lower bound, notice that each shortcut, as well as the root, makes at most $4r$ nodes susceptible to be part of a loop in $B_r^{(\overline{G_n})}(v_1)$ (see Figure 15) disregarding future shortcuts to be explored. Then,

³⁰Note that we did not explore the nodes they were rewired to since they are not connected to the connected component of the root in the i -th step of the breadth-first search process.

³¹Order decided based on the overall ordering of $B_r^{(\overline{G_n})}(v_1)$.

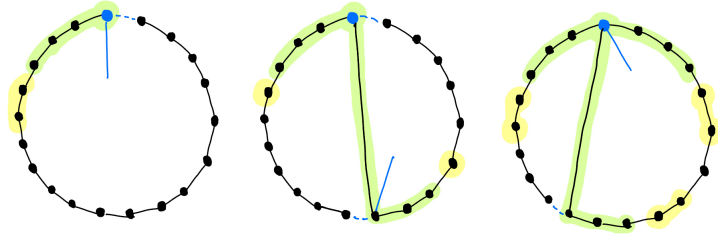


FIGURE 15. To avoid any loops in $B_r(o)$ for $r = 3$ where the blue node is the root in the above graphs, shortcuts in blue cannot be rewired to shaded nodes. Thus, we will shade at most $4r$ nodes around the root and at most another $4r$ nodes for each node explored through a shortcut in $B_r(o)$.

$$\begin{aligned}
 n_i &\geq n - \sum_{k=1}^{i-1} x_{k,4} \cdot (2(r - \text{dist}_g(v_k, v_1) - 1) + 1) + (1 + x_{k,1}) \cdot (2 \cdot 2(r - \text{dist}_g(v_k, v_1) - 1) + 1) \\
 &\geq n - \sum_{k=1}^{i-1} 4r \cdot (x_{k,4} + x_{k,1} + 1)
 \end{aligned} \tag{5.12}$$

dist_G denotes the length of the shortest path between nodes. We are subtracting 1 because we are interested in the distance of the root to the node w the shortcut of v_k leads to and adding the second one to count this node w . Secondly, we add 1 to $x_{k,1}$ to account for the root. We are multiplying only the second term in the sum by an extra 2 because mark (4) nodes do not have any mark (2) offspring due to rewiring but mark (1) nodes could have a mark (2) off-spring. All of this is to (over)count green-shaded nodes in Figure 15! Finally, we have another 2 because we don't want to rewire any two shortcuts to two nodes close enough to each other that the ring edges around them intersect and create a cycle (corresponding to (over)counting yellow shaded nodes in Figure 15). Similarly,

$$m_{i,k} \geq n_i - 4r \cdot (k - 1) \tag{5.13}$$

(3.4) and (3.5) give a lower bound (i.e. we are over-counting shaded nodes in Figure 15) because it is possible that due to randomness not all $4(r - \text{dist}_g(v_k, v_1) - 1)$ nodes on the ring around the node v_k would end up in $B_r^{(\overline{G_n})}(v_1)$ (e.g. some of the edges could be rewired *away* from the connected component).

We are ready to give the closed form!

$$\mathbb{P}(B_r^{(\overline{G_n})}(v_1) = \bar{t}) = \prod_{\substack{i \in [\bar{t}] \\ \text{dist}_G(v_0, v_i) < r}} \sum_{j=1}^4 \mathbb{1}_{\{M_n(v_i)=j\}} (1-\phi)^{x_{i,1}} \left(\phi \frac{n_i}{n} \right)^{x_{i,2}} (1-\phi)^{x_{i,3}} \phi^{1-x_{i,3}} g_j(x_{i,4}) \tag{5.14}$$

where

$$g(x_{i,4}) = \sum_{k=0}^{x_{i,4}} \mathbb{P}[\text{Bin}(l_i, \frac{1}{n}) = k, \text{no loop}] \mathbb{P}[\text{Bin}(s_i, \frac{\phi}{n}) = x_{i,4} - k, \text{no loop}] \tag{5.15}$$

Let's unpack this formula. the first $1 - \phi$ is the probability that the clock-wise edge does not get rewired (so it corresponds to mark (2) offspring) and $\phi \frac{n_i}{n}$ is the probability

that the clock-wise edge gets rewired (so it corresponds to mark (1) offspring) in a way that will not create a loop in the BFS, which is why we need to count n_i . Similarly, the second $1 - \phi$ is the probability that the anti-clock-wise edge does not get rewired (so it corresponds to mark (3) offspring) and ϕ is the probability that the anti-clock-wise edge gets rewired, so there is no type (3) offspring (*) (we don't care where these nodes are rewired since they are not part of the connected component containing the root so far). Lastly, $g(x_{i,4})$ is the probability that v_j has $x_{i,4}$ shortcuts rewired to it and they don't create any cycles in $B_r^{(\overline{G_n})}(v_1)$:

- $Bin(s_i, \frac{\phi}{n})$ is the number of shortcuts from non-explored nodes,
- $Bin(l_i, \frac{1}{n})$ is the number of shortcuts from anti-clockwise edges that were explored to be rewired but where they were rewired to has not been explored yet (recall (*)),
- and lastly, counting $m_{i,k}$ ensures they will not create a loop.

We will describe the limiting behavior of $g(x_{i,4})$. Again, the key point is that G_n is locally finite, so any finite sum of $x_{k,j}$ terms would be bounded. Then,

$$1 = \lim_{n \rightarrow \infty} \frac{n}{n} \geq \lim_{n \rightarrow \infty} \frac{n_i}{n} \geq \lim_{n \rightarrow \infty} \frac{n - \sum_{k=1}^{i-1} 4r \cdot (x_{k,4} + x_{k,1} + 1)}{n} = 1$$

and

$$1 = \lim_{n \rightarrow \infty} \frac{n}{n} \geq \lim_{n \rightarrow \infty} \frac{m_{i,k}}{n} \geq \lim_{n \rightarrow \infty} \frac{n_i - 4r \cdot (k - 1)}{n} = 1$$

By the Squeeze theorem, we conclude $\lim_{n \rightarrow \infty} \frac{n_i}{n} = \lim_{n \rightarrow \infty} \frac{m_{i,k}}{n} = 1$. We know that

$$\mathbb{P}[Bin(l_i, \frac{\phi}{n}) = k, \text{no loop}] \leq \mathbb{P}[Bin(l_i, \frac{\phi}{n}) = k] = \binom{l_i}{k} \left(\frac{1}{n}\right)^k \left(\frac{n-1}{n}\right)^{l_i-k} \rightarrow 0$$

except for $k = 0$. Second,

$$\mathbb{P}[Bin(s_i, \frac{\phi}{n}) = k, \text{no loop}] = \left(\prod_{k=0}^{x_{i,4}} \frac{m_{i,k}}{s_i}\right) \mathbb{P}[Bin(s_i, \frac{\phi}{n}) = k] \rightarrow \mathbb{P}[Poi(\phi) = k]$$

since $\frac{m_{i,k}}{s_i} \rightarrow 1$, $[Bin(s_i, \frac{\phi}{n}) = k] \rightarrow \mathbb{P}[Poi(\frac{s_i \phi}{n}) = k]$, and $\frac{s_i}{n} \rightarrow 1$. Hence,

$$g(x_{i,4}) \rightarrow \mathbb{P}[Poi(\phi) = x_{i,4}]$$

Finally, recall $\frac{n_i}{n} \rightarrow 1$. All in all, in the limit, we get

$$\mathbb{P}(B_r^{(\overline{G_n})}(v_1) = \bar{t}) \rightarrow \prod_{i \in [t]} \sum_{j=1}^4 \mathbb{1}_{\{M_n(v_i)=j\}} (1-\phi)^{x_{i,1}} (1-\phi)^{x_{i,3}} \phi^{1-x_{i,3}} \mathbb{P}[Poi(\phi) = x_{i,4}] \quad (5.16)$$

which is indeed same as $\mu(B_r^{(\overline{G})}(o))$ in (4.4). (The indicators of (4.4) in this latter formula are missing because they are trivially satisfied as $\mathbf{t}$ is taken from the limiting distribution.)

5.3.1. Second Moment. For the second moment, we are interested in calculating

$$\begin{aligned} \lim_{n \rightarrow \infty} \text{Var}(p_r^{\overline{G_n}}(t)) &= \lim_{n \rightarrow \infty} \mathbb{E}[p_r^{\overline{G_n}}(t)^2] - \mathbb{E}[p_r^{\overline{G_n}}(t)]^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{E}[(p_r^{\overline{G_n}}(t))^2] - \mathbb{P}(B_r^{(\overline{G_n})}(v_1) = \bar{t})^2 \end{aligned} \quad (5.17)$$

By symmetry of the neighborhood of the nodes, notice that

$$\begin{aligned} \mathbb{E}[p_r^{(\overline{G_n})}(t)^2] &= \mathbb{E}\left[\left(\frac{1}{n} \sum_{i \in [n]} \mathbb{1}(B_r^{(\overline{G_n})}(v_i) = \bar{t})\right)^2\right] = \frac{1}{n^2} \left[\left(n \cdot \mathbb{P}(B_r^{(\overline{G_n})}(u) = \bar{t}) \right) \right. \\ &\quad \left. + n \cdot (n-1) \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t}) \right] \end{aligned} \quad (5.18)$$

$$\begin{aligned} \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t}) &= \\ &= \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t} | \text{dist}_{G_n}(v, w) \leq 2r) \cdot \mathbb{P}(\text{dist}_{G_n}(v, w) \leq 2r) \\ &\quad + \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t} | \text{dist}_{G_n}(v, w) > 2r) \cdot \mathbb{P}(\text{dist}_{G_n}(v, w) > 2r) \end{aligned} \quad (5.19)$$

Given $\text{dist}_{G_n}(v, w) > 2r$, $B_r^{(\overline{G_n})}(v) = \bar{t}$ and $B_r^{(\overline{G_n})}(w) = \bar{t}$ are independent events. So, we get

$$\begin{aligned} \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t} | \text{dist}_{G_n}(v, w) > r) &\cdot \mathbb{P}(\text{dist}_{G_n}(v, w) > 2r) \\ &= \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t})^2 \cdot \mathbb{P}(\text{dist}_{G_n}(v, w) > 2r) \end{aligned}$$

Now, evaluating the former probability,

$$\begin{aligned} \mathbb{P}(\text{dist}_{G_n}(v, w) \leq 2r) &= \mathbb{P}(w \in B_{2r}^{(G_n)}(v) | v) = \mathbb{E}[\mathbb{1}_{\{w \in B_{2r}^{(G_n)}(v) | v\}}] \\ &= \mathbb{E}[\mathbb{E}[\mathbb{1}_{\{w \in B_{2r}^{(G_n)}(v) | v\}} | G_n]] \end{aligned}$$

where the last equality comes from the tower rule of expectations (here, it is important to recall that G_n is a random variable).

$$\mathbb{E}[\mathbb{E}[\mathbb{1}_{\{w \in B_{2r}^{(G_n)}(v) | v\}} | G_n]] = \mathbb{E}\left[\frac{|B_{2r}^{(G_n)}(v)| - 1}{n}\right] = \frac{\mathbb{E}[|B_{2r}^{(G_n)}(v)|] - 1}{n} \quad (5.20)$$

since node w was picked uniformly at random. Now, we will bound $\mathbb{E}[|B_{2r}^{(G_n)}(v)|]$.

$$\begin{aligned} \mathbb{E}[|B_{2r}^{(G_n)}(v)|] &= \mathbb{E}[\mathbb{E}[|B_{2r}^{(G_n)}(v)| | B_{2r-1}^{(G_n)}(v)]] \\ &= \mathbb{E}[\mathbb{E}[|B_{2r}^{(G_n)}(v)| | B_{2r-1}^{(G_n)}(v)] | B_{2r-2}^{(G_n)}(v)]] \\ &= \mathbb{E}[\mathbb{E}[|B_{2r}^{(G_n)}(v)| | B_{2r-1}^{(G_n)}(v)] | B_{2r-2}^{(G_n)}(v) | \dots | B_1^{(G_n)}(v)]] \dots \\ &\leq 1 + 2 + 4 + \dots + 2^{2r} = 2^{2r} - 1 \end{aligned} \quad (5.21)$$

given that each node has expected degree $1 + (1 - \phi) + (n \cdot \frac{\phi}{n}) = 2$. Substituting this upper bound back to 5.20, we get

$$\mathbb{P}(\text{dist}_{G_n}(v, w) \leq 2r) = \frac{\mathbb{E}[|B_{2r}^{(G_n)}(v)|] - 1}{n} \leq \frac{2^{2r} - 2}{n} \rightarrow 0 \quad (5.22)$$

since r is fixed. This means that

$$\mathbb{P}(\text{dist}_{G_n}(v, w) > 2r) = 1 - \mathbb{P}(\text{dist}_{G_n}(v, w) \leq 2r) \rightarrow 1$$

Combining all of these results with 5.19,

$$\mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t}, B_r^{(\overline{G_n})}(w) = \bar{t}) \rightarrow \lim_{n \rightarrow \infty} \mathbb{P}(B_r^{(\overline{G_n})}(v) = \bar{t})^2 \quad (5.23)$$

By equation 5.17 and 5.18, we conclude that

$$\begin{aligned}
\lim_{n \rightarrow \infty} \text{Var}(p_r^{(G_n)}(t)) &= \lim_{n \rightarrow \infty} \#(t)^2 \text{Var}(p_r^{(\overline{G_n})}(t)) \\
&\rightarrow \#(t)^2 \left[\frac{1}{n^2} \cdot n \mathbb{P}(B_r^{(\overline{G_n})}(u) = \bar{t})^2 \right. \\
&\quad \left. + \frac{1}{n^2} \cdot n(n-1) \mathbb{P}(B_r^{(\overline{G_n})}(u) = \bar{t})^2 - \mathbb{P}(B_r^{(\overline{G_n})}(u) = \bar{t})^2 \right] \\
&\rightarrow 0
\end{aligned}$$

Lastly, take any $\epsilon \geq 0$. By Chebychev's Inequality, we have

$$\mathbb{P}(|p_r^{(G_n)}(t) - \mathbb{E}[p_r^{(G_n)}(t)]| \geq \epsilon) \leq \frac{\text{Var}(p_r^{(G_n)}(t))^2}{\epsilon^2} \rightarrow 0$$

Thus, in the limit, we get

$$\lim_{n \rightarrow \infty} \mathbb{P}(|p_r^{(G_n)}(t) - \mu(B_r^{(G)}(o) \simeq h)| \geq \epsilon) = 0$$

which concludes the proof. $\square$

Pre-percolation, the small world network above survives almost surely since either mark (1) or mark (2) offspring will always exist at each tree level. By Theorem 5.10, the size of the whole graph would be an upper bound for the size of the giant, which is a trivial result. For the post-percolation case, let's consider the mean matrix of the multi-type branching process the small world network above converges to:

$$\begin{pmatrix}
\phi & 1-\phi & 1-\phi & \phi \\
\phi & 1-\phi & 0 & \phi \\
0 & 0 & 1-\phi & \phi \\
0 & 0 & 1-\phi & \phi
\end{pmatrix}$$

The maximum eigenvalue of this matrix is 1. Hence, by Theorem 4.15 and Theorem 4.19 the branching number of this tree as well as the critical percolation threshold is 1. If one thins the edges of this small world network with any positive probability $1-p$, since the branching number of the limit tree would be $p < 1$, the branching process will become extinct almost surely. Hence, by Theorem 5.10, there will be no giant with high probability in the small world network.

6. APPENDIX

Definition 6.1. (Martingale) A *filtration* on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a sequence $\{\mathcal{F}_n\}_{n=0}^\infty$ of σ -algebras of $\mathcal{F}$ such that:

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}.$$

This increasing sequence represents the information available up to time n . Hence, any event known at time n is also known at all subsequent times $m \geq n$.

Now, let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_n\}_{n=0}^\infty$ a filtration of $\mathcal{F}$. A sequence of random variables $\{X_n\}_{n=0}^\infty$, adapted to $\{\mathcal{F}_n\}_{n=0}^\infty$, is called a ***martingale*** if:

- (1) X_n is $\mathcal{F}_n$ -measurable for each n .
- (2) Each X_n is integrable, i.e., $\mathbb{E}[|X_n|] < \infty$.
- (3) For all $n \geq 0$, the conditional expectation satisfies

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n.$$

We call this sequence a **supermartingale** if $E[X_{n+1}|\mathcal{F}_n] \leq X_n$ almost surely for all $n \geq 0$ instead of (3) above.

Finally, we say a martingale $(M_n)_{n \geq 0}$ is in L^p , if each random variable M_n in the sequence satisfies

$$\mathbb{E}[|M_n|^p] < \infty$$

for every n and some fixed $p \geq 1$.

Theorem 6.2. (*Martingale Convergence Theorem*) *Let X be a supermartingale bounded in L^1 . Then X_n converges and is finite almost surely. Moreover, let $X_\infty = \liminf_{n \rightarrow \infty} X_n$, then $X_\infty \in \mathcal{F}_\infty$ and $E[|X_\infty|] < +\infty$.*

There are stronger forms of convergence which we define next.

Definition 6.3. A sequence of random variables $(X_n)_{n \geq 1}$ converges in L^p to a random variable X if:

$$\lim_{n \rightarrow \infty} \mathbb{E}[(X_n - X)^p] = 0.$$

Convergence in L^1 is also known as convergence in mean since it means that the expectation of the random variable converges. Conveniently, we know that convergence in L^m implies convergence L^n whenever $m \geq n$!

Now, we give an example that shows that the conditions of the Martingale Convergence Theorem do not guarantee convergence in mean. Let $\{S_n\}$ be a simple random walk on the integers started at 1 and

$$T = \inf\{n > 0 : S_n = 0\}.$$

So, one can view $\{S_{T \wedge n}\}$ as a martingale that is stopped once S_n hits 0. Then $\{S_{T \wedge n}\}$ is a nonnegative martingale. It can only converge to 0. (Any other integer value would not be possible because convergence would have to have occurred at a finite time and the next time step would have to be different.) But $E[X_0] = 1 \neq 0$ [7]. Then, when does a martingale converge in mean?

Theorem 6.4. (*Martingale Convergence in Mean*) *Let M be a martingale with $M_n \in L^2$. Then M is bounded in L^2 if and only if*

$$\sum_{k \geq 1} \mathbb{E}[(M_k - M_{k-1})^2] < +\infty.$$

When this is the case, M_n converges almost surely (a.s.) and in L^2 and hence in L^1 as well! (There are weaker conditions for L^1 convergence but for this exposition, this definition suffices.)

Theorem 6.5. (*Fatou's Lemma*) *Let $(f_n)_{n=1}^\infty$ be a sequence of non-negative measurable functions on a measure space $(X, \mathcal{M}, \mu)$. Then,*

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

Viewing a sequence of non-negative random variables $(X_n)_{n=1}^\infty$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ as functions from this space to positive reals and the expectation as an integral with respect to a probability measure, Fatou's Lemma asserts that:

$$\mathbb{E}[\liminf_{n \rightarrow \infty} X_n] \leq \liminf_{n \rightarrow \infty} \mathbb{E}[X_n].$$

We have a similar theorem for random variables in general.

Theorem 6.6. (*Dominated Convergence Theorem*) If $\{f_n\}$ is a sequence of measurable functions converging pointwise to a function f almost surely and there exists an integrable function g (i.e. $\int g d\mu < \infty$) such that $|f_n(\cdot)| \leq g(\cdot)$ for all n almost surely, then

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0,$$

and consequently,

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

which gives us a way to interchange limits and integrals. The integral would give an expectation if f_n is a random variable with measure μ . Hence, as long as random variable $X_n \rightarrow X$ almost surely, satisfies $|X| < Y$ almost surely, and $\mathbb{E}[Y] < \infty$, we have $\mathbb{E}[X_n] = \mathbb{E}[\lim_{n \rightarrow \infty} X]$.

Theorem 6.7. (*Fubini's Theorem*) If either $\int_A \int_B |f(x, y)| dy dx$ or $\int_B \int_A |f(x, y)| dx dy$ is finite, then

$$\int_A \left(\int_B f(x, y) dy \right) dx = \int_B \left(\int_A f(x, y) dx \right) dy = \int_{A \times B} f(x, y) d(x, y)$$

As an application of this theorem, the following proof of the *Law of Total Expectations* uses Fubini's Theorem in the second line to be able to swap the order of integrals:

$$\begin{aligned} \mathbb{E}[X] &= \int_x x f_X(x) dx = \int_x \int_y f_{X,Y}(x, y) dy dx \\ &= \int_x \int_y f_{X|Y}(x, y) f_Y(y) dy dx = \int_y \int_x f_{X|Y}(x, y) f_Y(y) dx dy \\ &= \int_y \int_x f_{X|Y}(x, y) dx f_Y(y) dy = \int_y \mathbb{E}[X|Y] f_Y(y) dy = \mathbb{E}[\mathbb{E}[X|Y]] \end{aligned}$$

One can swap the integrals above if Fubini's conditions for the inner integral hold and if the densities of X and Y exist.

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